

Yes/No Questions Across Chinese Varieties and English

Abstract

This article provides a morphosyntactic and semantic analysis of ten yes/no question constructions across English and Chinese varieties (e.g., Mandarin, Yixing, Nanjing, Suzhou, Hefei, Kunming, Liangshan, many of which are understudied), and shows that the apparently complex crosslinguistic variation is in fact complete systematic variation generated by just three binary parameters. The first parameter is whether or not the question involves a polar Disjunction Phrase (DisjP) in syntax. The questions with a polar DisjP vary in two more parameters: whether there is overt or covert *wh*-movement, and whether the DisjP is realised as a single Vocabulary Item or multiple. Our study has two theoretical implications. The first is for question semantics and typology: we provide syntactic and morphological evidence for two distinct question operators (or interrogative C⁰s), one that takes a singleton set of propositions (Hamblin 1973) and the other that takes a proposition set with multiple members (e.g., Kratzer and Shimoyama 2002); we also argue that the yes/no questions without DisjP do not involve any *wh*-dependency, and thus no (c)overt *wh*-movement or even genuine *wh*-in-situ, thus enriching the typology of questions. Our second theoretical contribution lies in theories of parametric variation. The three parameters proposed in this article have three different sources, all of which should be taken into account in parametric theories: linearization, phonological realization of features during Lexical Insertion, and the (un)availability of features in the lexicon.

Keywords: yes/no questions; polar Disjunction Phrase; *wh*-movement; parametric variation; Chinese; English

1. Introduction

Yes/no questions (i.e., questions that can be answered by ‘yes’ or ‘no’) uniquely sit at the intersection of two important topics in linguistic theory—alternatives and questions. There is variation in both the presentation of alternatives and question formation, leading us to expect wide variation in yes/no question formation. Within a language, we may see yes/no questions differ in whether they involve polarity disjunction: for example, in English, some yes/no questions involve a polarity disjunction phrase (*or not*), while others don’t. Across languages, there is well-known binary variation in *wh*-question formation: the *wh*-phrase moves overtly in some languages but covertly in others. Since these two properties (i.e. alternatives and questions) intersect in yes/no questions, we may expect yes/no questions to have rich variation within and across languages. This paper examines this rich variation in yes/no questions in English and the Chinese language family (including Mandarin and other varieties), and shows that they have the complete parametric variation we expect from the paradigm. Furthermore, they vary in a third dimension: if the yes/no question involves a polarity disjunction, whether that disjunction is realised as a single Vocabulary Item (VI) or multiple. Thus, our morphosyntactic and semantic theories of yes/no questions allow us to find order in the apparently noisy data: the apparently diverse yes/no question types across English and the Chinese varieties are in fact complete systematic variation generated by just three binary parameters. This is significant for the understanding of question semantics and typology, where we argue

that there are at least three types of questions and two types of question operators (or interrogative C⁰s). Our work also bears on parametric variation in Minimalism by arguing that a combination of two major approaches, the Borer-Chomsky Conjecture (Baker 2008; Borer 2013; Roberts 2019) and the externalisation approach (Berwick and Chomsky 2016; Chomsky 2019; Chomsky et al. 2023) together accounts for the variation of yes/no questions we observe; additionally, they interact with each other to generate the apparent complexity of variation.

We first introduce the theoretical background on each binary variation using basic data, beginning with polarity disjunction. In English, yes/no questions may or may not involve *or not*:

- (1)a. Does John like coffee or not?
- b. Does John like coffee?
- (2)a. I wonder whether John likes coffee or not.
- b. I wonder whether John likes coffee.

Contrary to the claim that these two types of yes/no questions are equivalent, and there is silent *or not* when we don't hear it (e.g., Larson 1985), it has been observed that these two types of yes/no questions sometimes have different distributions (e.g., Karttunen 1977a; Bolinger 1978; Huddleston 1994; Eckardt 2007; Rawlins 2008; Biezma 2009; Biezma and Rawlins 2012). For example, Karttunen observed that only yes/no questions without *or not* can be the complement of *doubt*:

- (3)a. #I doubt whether John likes coffee or not. *ALTQVN*
- b. I doubt whether John likes coffee. *POLQ*

Evidence like this has led to the proposal that there are two different types of yes/no questions: alternative questions with a disjunction phrase (DisjP) *or not* that contrasts in polarities ((1a) and (2a), which we call *ALTQVN* following Biezma and Rawlins 2012); and questions without any DisjP ((1b) and (2b), which we call *POLQ*) (e.g., Hamblin 1973; Karttunen 1977b; Rawlins 2008; Biezma 2009; Pruitt and Roelofsen 2011; Biezma and Rawlins 2012; Farkas and Roelofsen 2012; Uegaki 2023).

There are two different types of approaches to the contrast between (3a&b). The first was that the *ALTQVN* in (3a) denotes a doubleton set {John likes coffee, John doesn't like coffee}, whereas the *POLQ* in (3b) denotes a singleton set {John likes coffee}, just like a declarative *that*-clause, and *doubt* selects for singleton sets (Biezma and Rawlins 2012). The second approach adopted the Hamblin semantics of questions, according to which both the *ALTQVN* and the *POLQ* denote a doubleton set, but the *POLQ* highlights only one proposition {John likes coffee}, whereas the *ALTQVN* highlights two {John likes coffee, John doesn't like coffee} (Pruitt and Roelofsen 2011; Farkas and Roelofsen 2012; Uegaki 2023). *Doubt* requires its complement to highlight exactly one proposition.

The second approach has an important consequence: there are two different interrogative C^0 s, occurring in *ALTQVNs* and *POLQs* respectively. This is based on Hamblin disjunction semantics, where the TP in *ALTQVNs* denotes a doubleton set (i.e., {John likes coffee, John doesn't like coffee} in (3a); Alonso-Ovalle 2005; Simons 2005; Alonso-Ovalle 2006), whereas the TP in *POLQs* denotes a singleton set (i.e., {John likes coffee} in (3b)). To derive the doubleton meaning of these questions, we thus need two different interrogative C^0 s: the C^0 in *ALTQVNs* is an identity function (same as the C^0 in

wh-constituent questions), whereas the C^0 in POLQs turns a singleton set into a doubleton. This proposal is based on English, where the two interrogative C^0 s look the same. This drew criticism about this approach—that is, across languages, we do not see morphological evidence of two different C^0 s (Biezma and Rawlins 2015).

This paper argues for two distinct interrogative C^0 s with a close cross-dialectal examination of two Chinese varieties, where the two C^0 s are realised differently. We will not only use the ‘doubt’ test, but also develop new diagnostics to show that two apparently similar yes/no question constructions in the Chinese varieties are in fact ALTQVNs and POLQs respectively. This is the first binary parameter we investigate.

Specifically, we will contrast *fe/me*-marked questions (hereafter *fe/me*-questions) in Chinese Yixing dialect (4a) with *ma*-marked questions (hereafter *ma*-questions) in Mandarin (4b).

- | | | | | | | |
|-------|---------------------------------------|--------|-------|--------|-----|-------------------------------|
| (4)a. | Zangsa | huexi | qie | kafi | fe? | Yixing <i>fe/me</i> -question |
| | Zhangsan | like | drink | coffee | FE | |
| | ‘Does Zhangsan like to drink coffee?’ | | | | | |
| b. | Zhangsan | xihuan | he | kafei | ma? | Mandarin <i>ma</i> -question |
| | Zhangsan | like | drink | coffee | MA | |
| | ‘Does Zhangsan like to drink coffee?’ | | | | | |

These two types of yes/no questions appear to be very similar because they are both marked by a sentence-final particle—*fe* (or *me*, depending on the aspect of the clause), and *ma*. But we will show that Mandarin *ma*-questions are POLQs and do not involve a DisjP

in the syntax, whereas Yixing *fe/me*-questions are ALTQVNs and involve a DisjP. The C⁰ in ALTQVNs is null in Yixing and Mandarin, whereas the C⁰ in POLQs is *ma* in Mandarin.

ALTQVNs can be analysed on a par with general alternative questions such as (5). The only difference between them is that the DisjP in ALTQVNs disjoins the positive and negative polarities, whereas the DisjP in alternative questions may disjoin other constituents.

(5) I wonder whether John likes coffee or tea.

If we follow Larson (1985), Romero and Han (2004), Wu (2022), and others' view that alternative questions and ALTQVNs involve a *wh*-element (e.g., *whether* in English) that moves from the left edge of the DisjP to Spec, CP, then we can unify alternative questions with *wh*-constituent questions. This leads us to ask whether the yes/no questions in Chinese are *wh*-questions. We will show that the ALTQVNs (i.e., Yixing *fe/me*-questions) have *wh*-properties, but POLQs (i.e., Mandarin *ma*-questions) don't. This has the consequence that the C⁰ in ALTQVNs triggers *wh*-movement, whereas the C⁰ in POLQs doesn't.

The lack of *wh*-movement in *ma*-questions might remind us of the lack of *wh*-movement in genuine *wh*-in-situ constructions (e.g., English superiority-violating questions; Pesetsky 2000; Kotek 2018), where a *wh*-in-situ question has *wh*-question meaning via alternative computation, unselective binding or choice functions, but we will show that *ma*-questions do not involve genuine *wh*-in-situ (i.e., they are not subject to intervention effects). Our work on *ma*-questions thus enriches the typology of questions, suggesting that there are at least three types of questions: (a) questions with (overt or

covert) *wh*-movement (e.g., standard *wh*-constituent-questions, Yixing *fe/me*-questions and Mandarin *bu/mei*-questions); (b) genuine *wh*-in-situ questions without any *wh*-movement (e.g., English superiority-violating questions and Japanese *wh*-in-situ; Kotek 2018); and (c) yes/no questions without *wh*-movement or *wh*-in-situ (e.g., Mandarin *ma*-questions).

We will not only argue that Yixing *fe/me*-questions involve *wh*-movement, but also that this movement is overt. This may be surprising because *wh*-constituent movement in Chinese has traditionally been assumed to be covert by default since C.-T. J. Huang (1982). But we will show that not all yes/no questions in Chinese involve overt *wh*-movement; some involve covert *wh*-movement as well, such as A-not-A and *shifou*-questions in Mandarin and *a/ke*-questions in other Chinese varieties. This is the second parameter of variation—if a yes/no question involves *wh*-movement, whether this movement is overt or covert. A-not-A questions, for example, typically reduplicate the first syllable of the main predicate and insert negation after the reduplicant:

(6) Zhangsan xi bu xihuan he kafei?

Zhangsan RED NEG like drink coffee

‘Does Zhangsan like to drink coffee?’

We will argue that A-not-A questions involve a DisjP, just like *fe/me*-questions, but their DisjP (i.e., RED NEG) appears sentence-medially, rather than at the clause edge as in *fe/me*-questions. We still observe effects of cross-clausal movement in A-not-A questions, suggesting that they involve covert *wh*-movement, in support of C.-T. J. Huang (1991).

Having introduced two parameters underlying the cross-linguistic variation of yes/no questions—whether they involve a DisjP, and whether they involve overt or covert *wh*-movement—the third parameter is if they involve a DisjP, whether that DisjP is phonologically realised as a single Vocabulary Item in the sense of Distributed Morphology (Halle and Marantz 1993; Marantz 2006; Embick 2010; 2015 a.o.) or as multiple VIs. We will argue that Mandarin A-not-A, *shifou* and Yixing *fe/me* realise the DisjP as multiple VIs. They will be contrasted with Nanjing, Suzhou (including Taicang and Changshu), Hefei, and Kunming varieties, whose yes/no questions insert a single particle such as *a* (in Nanjing and Suzhou) and *ke* (in Hefei and Kunming) before the VP (hereafter *a/ke*-questions):

(7) Zhangsan *a* xihuan huo kafei? *Nanjing*

Zhangsan *A* like drink coffee

‘Does Zhangsan like to drink coffee?’

We will argue that *a* and *ke* are the DisjP in these questions, and a single VI. We will further show that the interaction between the second and third parameters predicts the existence of another yes/no question construction, where the DisjP is realised as a single VI like *a/ke*, and it undergoes overt *wh*-movement as in Yixing *fe/me*-questions. These are precisely Liangshan *bo*-questions, whose question marker *bo* is a single VI realising DisjP and undergoes overt *wh*-movement to the sentence-final position.

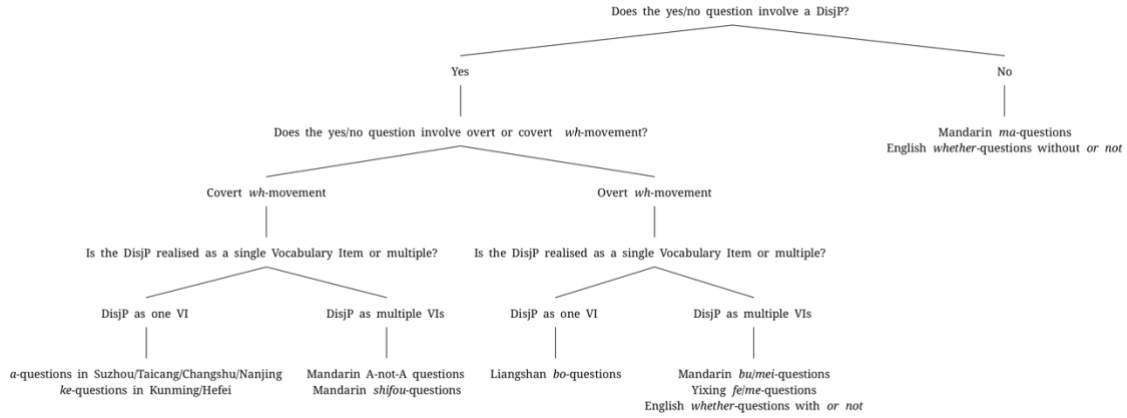
(8) Zhangsan xihuan he kafei bo? *Liangshan*

Zhangsan like drink coffee BO

‘Does Zhangsan like to drink coffee?’

Below is a summary of the three binary parameters of yes/no questions that we will examine in detail in the remainder of this paper.¹

(9) Parameters of Yes/no Questions



We demonstrate the rich diversity of yes/no questions in the Chinese varieties and English using Roberts' (2019) parameter hierarchy. With only three binary parameters at play, we observe complex but systematic cross-linguistic variation of yes/no questions. Seemingly noisy data can be reduced to a finite number of parameters, consonant with parametric variation in the Minimalist Program.

¹ Some might assume implicitly that there can only be parametric variation across different languages. But some parameters in our study describe variation within a language: for example, Mandarin and English have both ALTQVNs and POLQs, thus our first parameter is an intra-linguistic one. But some versions of the Borer-Chomsky Conjecture (e.g., Chomsky 2001) would allow parametric variation within a language, since they claimed that variation is due to different features carried by a lexical item. Previous studies have also implied that there can be parametric variation within a certain language, such as the co-existence of overt *wh*-movement, covert *wh*-movement and genuine *wh*-in situ in English (Pesetsky 2000; Kotek 2016).

The organisation of this paper follows the parametric hierarchy in (9) closely. We begin with a discussion of the first parameter (i.e., whether a yes/no question involves a DisjP or not, making it a POLQ or an ALTQVN) in section 2, where we present our core assumptions about the syntax of yes/no questions and argue that Yixing *fe/me*- and Mandarin *bu/mei*-questions are ALTQVNs, whereas Mandarin *ma*-questions are POLQs. We will also discuss the implications this has for the typology of question operators and that of questions in general. Section 3 moves on to the second parameter (i.e., whether a yes/no question involves overt or covert *wh*-movement), and proposes a new analysis of A-not-A questions that differs from but draws insights from the main approaches in the literature, Zhu (1985); C.-T. J. Huang (1991); C.-T. J. Huang et al. (2009). Section 4 examines the third parameter (i.e., whether the DisjP is realised as a single VI or multiple) based on Nanjing, Suzhou, Hefei, and Kunming. This section also shows that an interaction of the parameters makes an important prediction that is borne out in another Chinese variety, Liangshan. Section 5 concludes with a discussion of the two main theoretical contributions of the paper: one for question semantics and typology, and the other for parametric theories. Specifically, our parametric hierarchy supports three types of theories of parametric variation in the Minimalist Program, with each parameter supporting one type of theory.

2. Whether the yes/no question involves a DisjP

This section focuses on three types of yes/no questions in the Chinese family, including two in Mandarin and one in Yixing Chinese, a variety of Chinese Wu spoken in Jiangsu Province, China. While all three types insert a sentence-final particle (SFP) at the end of

a declarative, we will show that they represent two different ways of deriving a yes/no question, with and without a DisjP. We begin this section by first comparing two of these three types of questions, Mandarin *ma*-questions and Yixing *fe/me*-questions in section 2.1, and argue that the former don't involve a DisjP, but the latter do. Then we provide their syntactic analyses in section 2.2, and argue that Mandarin *ma*-questions involve a movement from C⁰ to SpeechAct⁰ in section 2.3. In section 2.4 we add the third type of question, Mandarin *bu/mei*-questions, to the discussion, which we argue are a hybrid of *ma*- and *fe/me*-questions.

2.1. Yixing *fe/me*-questions involve a polar DisjP, but Mandarin *ma*-questions don't

In Mandarin Chinese, a yes/no question can be created by adding the SFP *ma* to the end of a declarative sentence ((4b) is repeated below):

(10) Zhangsan xihuan he kafei.

Zhangsan like drink coffee

‘Zhangsan likes to drink coffee.’

(4)b. Zhangsan xihuan he kafei ma?

Zhangsan like drink coffee MA

‘Does Zhangsan like to drink coffee?’

In Yixing, a yes/no question can be created by adding *fe* or *me* to the end of a declarative sentence. *Me* is used in sentences with a perfective viewpoint aspect (i.e. sentences that ‘present a situation as a single whole’; Smith 1997:103), and *fe* is used elsewhere. An

additional SFP *a* can follow them, and can fuse with *fe* or *me* to turn into *fa* and *mang* respectively.

(11) Zangsa huexi qie kafi de.

Zhangsan like drink coffee DE

‘Zhangsan likes to drink coffee.’

(12) Zangsa huexi qie kafi {fe/fe a/fa}?

Zhangsan like drink coffee FE/FE A/FA

‘Does Zhangsan like to drink coffee?’

(13) Zangsa jinzaio qie kafi ge.

Zhangsan today drink coffee GE

‘Zhangsan drank coffee today.’

(14) Zangsa jinzaio qie kafi {me/me a/mang}?

Zhangsan today drink coffee ME/ME A/MANG

‘Did Zhangsan drink coffee today?’

We think that the sentence-final *fe* and *me* in yes/no questions are genuine negations for two reasons. First, they are identical with the negation words in the language. These negation words are also sensitive to the aspect of the sentence: *me* is the negation used with the perfective aspect, and *fe* is the negation elsewhere:

(15) a. Zangsa fe huexi qie kafi.

Zhangsan FE like drink coffee

‘Zhangsan doesn’t like to drink coffee.’

b. Zangsa jinzao me qie kafi.

Zhangsan today ME drink coffee

‘Zhangsan didn’t drink coffee today.’

Second, like negation, the SFPs *fe* and *me* are incompatible with focus and commitment particles like *de* and *lie*. We do not provide an analysis of *de* and *lie*, but just note here that *fe* and *me* have the same distribution as negation with respect to *de* and *lie*. *De*² or *lie* is required in positive declaratives but disallowed in negative ones:

(16) a. Zangsa huexi qie kafi de.

Zhangsan like drink coffee DE

‘Zhangsan likes to drink coffee.’

b. Zangsa qie-le sa bai kafi lie.

Zhangsan drink-PERF three cup coffee LIE

‘Zhangsan has drunk three cups of coffee.’

(17) a. Zangsa fe huexi qie kafi (*de).

Zhangsan NEG like drink coffee DE

‘Zhangsan doesn’t like to drink coffee.’

b. Zangsa me qie-le sa bai kafi (*lie).

Zhangsan NEG drink-PERF three cup coffee LIE

‘Zhangsan didn’t drink three cups of coffee.’

² *De* in Yixing is not equivalent to the particle *de* in Mandarin. The Yixing particle corresponding to the Mandarin *de* is *ge*, as in (13).

De and *lie* are disallowed in *fe/me*-questions; instead, only *a* can co-occur with *de* and *lie*.

We take the same distribution of *fe/me* and negation to indicate that *fe* and *me* are negation.

- (18) a. Zangsa zenglao huexi qie kafi de { *fe/*fe a/*fa/a } ?
 Zhangsan really like drink coffee DE FE/FE A/FA ?
 ‘Is it true that Zhangsan really likes to drink coffee?’
- b. Zangsa yijing qie-le sa bai kafi lie { *me/*me a/*mang/a } ?
 Zhangsan already drink-PERF three cup coffee LIE ME/ME A/MANG/A
 ‘Did Zhangsan already drink three cups of coffee?’

While Mandarin *ma* is not currently used as negation in the language, there is evidence that it evolved from negation historically (e.g., Aldridge 2011). This makes Mandarin *ma*-questions and Yixing *fe/me*-questions look very similar: they are all created by inserting (what is currently or historically) a negation particle at the end of a declarative. But we will argue using four diagnostics that they are very different: Yixing *fe/me*-questions are ALTQVNS because they involve a polar DisjP equivalent to English *or not*, whereas Mandarin *ma*-questions are POLQs because they don’t involve any DisjP or *wh*-movement. As we present those four tests in Mandarin and Yixing, we will also use them on English *whether*-questions to show that *whether*-questions with overt *or not* are ALTQVNS, whereas those without overt *or not* are POLQs.

First, *ma*-questions can be answered with *shi de*, a cleft construction that we translated as ‘It is so’ (C.-T. J. Huang et al. 2009), whereas *fe/me*-questions can’t. Instead, the verb is required in the polar answer.

(19) A: Zhangsan xihuan he kafei ma?

Zhangsan like drink coffee MA

‘Does Zhangsan like to drink coffee?’

B: shi de.

COP DE

‘It is so.’

(20) A: Zhangsan huexi qie kafi {fe/fe a/fa}?

Zhangsan like drink coffee FE/FE A/FA

‘Does Zhangsan like to drink coffee?’

B1: #si ge.

COP GE

‘It is so.’

B2: huexi de.

like DE

‘Yes.’

Following Roelofsen and van Gool (2010), Pruitt and Roelofsen (2011), Roelofsen and Farkas (2015), and Mohammadi and Romero’s (2024) analysis of ‘yes’ and ‘no’ answers in English and Farsi, we think that the ‘so’-answer requires the question to highlight only one proposition. According to these proposals, the highlighted value of a question only includes the answers that are mentioned overtly in the syntactic form of the TP. Thus, POLQs highlight only one proposition (i.e., the TP), whereas ALTQVNs highlight two propositions at the same time (i.e., the positive proposition and the negative). We thus

take the contrast between (19) and (20) to indicate that *fe/me*-questions involve a DisjP, whereas *ma*-questions don't.

In parallel, English questions with *or not* cannot be answered using *so*, whereas those without *or not* can, suggesting that the former are ALTQVNs and the latter are POLQs:³

(21) A: Does John like to drink coffee or not? ALTQVN

B: #I guess so / #It seems so / #Maybe so.

(22) A: Does John like to drink coffee? POLQ

B: I guess so / It seems so / Maybe so.

The second diagnostic is based on Karttunen's (1977a) observation in English that *doubt* can only take *that*-declaratives and yes/no questions without *or not*, but not yes/no questions with *or not*:

(23) a. John doubts that Marie stole the money.

b. John doubts whether Marie stole the money. POLQ

c. #John doubts whether Marie stole the money or not. ALTQVN

As discussed in section 1, we follow Pruitt and Roelofsen (2011), Farkas and Roelofsen (2012) and Uegaki (2023) in assuming that *doubt* requires its complement to highlight exactly one proposition. POLQs highlight only one proposition, whereas ALTQVNs highlight two, the positive and the negative propositions.

³ A reviewer pointed out that (22) cannot be answered by *So he does*. We think this might have to do with the reactive nature of *So he does* (which can be used in *John's mother asked him to take out the trash, and so he did*), and leave this question to future research.

In parallel, ‘doubt’ in Mandarin (or better translated as ‘suspect’) and Yixing can only take a single proposition (24), not a disjunction of two propositions (25).⁴ There are two ‘or’s in Mandarin, *haishi*, the ‘or’ associated with ‘whether’, and *huozhe*, the ‘or’ associated with ‘either’. *Haishi* is not possible in the clause embedded by ‘doubt’ (25), but *huozhe* is (26). Crucially, (26) only has the reading that Zhangsan suspects that one of Lisi and Wangwu stole the money.

(24) Zhangsan huaiyi Lisi tou-le qian.

Zhangsan suspect Lisi steal-PERF money

‘Zhangsan suspects Lisi stole the money.’

(25) #Zhangsan huaiyi Lisi haishi Wangwu tou-le qian.

Zhangsan suspect Lisi or_{whether} Wangwu steal-PERF money

Intended: ‘Zhangsan suspects whether Lisi or Wangwu stole the money.’

(26) Zhangsan huaiyi Lisi huozhe Wangwu tou-le qian.

Zhangsan suspect Lisi or_{either} Wangwu steal-PERF money

‘Zhangsan suspects that either Lisi or Wangwu stole the money.’

‘Doubt’ cannot embed *fe/me*-questions (27), even though *fe/me*-questions can generally be embedded (e.g. under ‘know’, (28)). This suggests that *fe/me*-questions are ALTQVNS. It is worth mentioning that while *fe/me*-questions can be generally embedded, those with the additional *a* can’t be, a fact we will discuss in depth later in section 2.3.

⁴ *Huaiyi* actually has the meaning of English *suspect*, not *doubt* because the subject of *huaiyi* is inclined to believe that the complement proposition of *huaiyi* is true, not false. But since *huaiyi* requires highlighting a single proposition, we take it to behave like *doubt* in this aspect.

(27) #Zangsa huani Lisi huexi qie kafi {fe/fe a/fa}.⁵

Zhangsan suspect Lisi like drink coffee FE/FE A/FA

Intended: ‘Zhangsan suspects whether Lisi likes to drink coffee.’

(28) wo xiaozhe lao Zangsa huexi qie kafi {fe/*fe a/*fa}.

I know LAO Zhangsan like drink coffee FE/FE A/FA

‘I know whether Zhangsan likes to drink coffee.’

‘Doubt’ cannot embed *ma*-questions, either, which may at first appearance suggest that *ma*-questions involve a DisjP:

(29) *Zhangsan huaiyi Lisi xihuan he kafei ma.

Zhangsan suspect Lisi like drink coffee MA

Intended: ‘Zhangsan suspects whether Lisi likes to drink coffee.’

But we think this is because *ma*-questions cannot be embedded at all, even under verbs that can otherwise embed questions, such as ‘know’:

(30) *wo zhidao Lisi xihuan he kafei ma.

I know Lisi like drink coffee MA

Intended: ‘I know whether Lisi likes to drink coffee.’

Thus, *fe/me*-questions cannot be embedded under ‘doubt’ because they are ALTQVNs and involve a DisjP, whereas Mandarin *ma*-questions and Yixing *a*-questions cannot be

⁵ This sentence can be interpreted as a yes/no question, with both these meanings ‘Does Zhangsan suspect that Lisi likes to drink coffee, or does he suspect that Lisi doesn’t like to drink coffee?’ and ‘Does Zhangsan suspect that Lisi likes to drink coffee, or does he not suspect that Lisi likes to drink coffee?’ We will discuss these interpretations in section 2.2.

embedded in general. Section 2.3 will provide an account of the embeddability of *fe* and *me* but the unembeddability of *ma* and *a*.

Our third diagnostic is based on Rawlins' (2008) observation that English ALTQVNs can be used as unconditionals, whereas POLQs cannot:

- (31) a. #Whether John likes to drink coffee, we will meet at Starbucks. *POLQ*
 b. Whether John likes to drink coffee or not, we will meet at Starbucks. *ALTQVN*

In parallel, *fe/me*-questions can occur in unconditionals, suggesting they are ALTQVNs.

Sentence-final *a* cannot occur here, which again we will explain in section 2.3.

- (32) fegue ni mingzao lai wo guo qie va {fe/*fe a/*fa},
 no.matter you tomorrow come my home have meal FE/FE A/FA
 wo jinzao zongguai yao ma maduo cai de.
 I today anyway will buy many groceries DE
 'Whether you're coming for a meal or not, I'll buy many groceries today anyway.'

Ma-questions cannot be embedded in unconditionals, but this does not indicate they are POLQs because as we saw in (30), *ma*-questions are generally unembeddable.

- (33) *buguan ni mingtian lai wo jia chi fan ma,
 no.matter you tomorrow come my home have meal MA
 wo jintian dou yao mai henduo cai de.
 I today DOU will buy many groceries DE

Intended: 'Whether you're coming for a meal or not, I'll buy many groceries.'

Our final diagnostic of a polar DisjP involves negative yes/no questions. *Ma*-questions can involve a negative proposition (C.-T. J. Huang et al. 2009), but *fe/me*-questions cannot. Instead, only *a* appears in negative yes/no questions in Yixing:

(34) a. Zhangsan bu he kafei ma?

Zhangsan NEG drink coffee MA

‘Does Zhangsan not drink coffee?’

b. Zhangsan mei he kafei ma?

Zhangsan NEG drink coffee MA

‘Did Zhangsan not drink coffee?’

(35) a. Zangsa fe qie kafi { *fe/*fe a/*fa/a }?

Zhangsan NEG drink coffee FE/FE A/FA/A

Intended: ‘Does Zhangsan not drink coffee?’

b. Zangsa me qie kafi { *me/*me a/*mang/a }?

Zhangsan NEG drink coffee ME/ME A/MANG/A

Intended: ‘Did Zhangsan not drink coffee?’

These negative yes/no questions check the truth and falsity of a negative proposition and can license negative polarity items (NPIs), and therefore have what Romero and Han (2004) called the inner negation reading, equivalent to their English translations. Like their English translations, they can involve bias, but do not have to. We observe that in parallel, English POLQs allow negative propositions, whereas ALTQVNs don’t:

(36) a. *Does John not drink coffee or not?

b. Does John not drink coffee?

We therefore think that the yes/no question's compatibility with a negative proposition can be a diagnostic of whether it involves a polar DisjP: Yixing *fe/me*-questions do, but Mandarin *ma*-questions don't.

To sum up, we have presented four tests, answerability by 'so', embeddability by 'doubt', embeddability in unconditionals, and negative yes/no questions, that all suggest that Yixing *fe/me*-questions are ALTQVNs and involve a polar DisjP, whereas Mandarin *ma*-questions are POLQs and don't involve any polar DisjP.

It is worth mentioning that Mandarin and Yixing have yes/no questions with a transparent DisjP disjoined by Mandarin *haishi* or Yixing *asi* 'or_{whether}'. 'Or_{whether}' can disjoin phrases of various sizes, such as DPs, VPs and clauses, and 'or_{whether}' can be silent. When predicates of opposite polarities are disjoined, we get a yes/no question meaning:

(37) Zhangsan he kafei (haishi) bu he?

Zhangsan drink coffee or_{whether} NEG drink

'Does Zhangsan drink coffee or not?'

These questions pass all our tests of polar DisjP, and since they clearly involve a DisjP, we will not discuss them further. Section 2.2.3 will argue that *fe/me*-questions are not derived from these transparently alternative questions plus ellipsis.

2.2. Syntax of *fe/me*- and *ma*-questions

Having shown that *fe/me*-questions are ALTQVNs but *ma*-questions are POLQs, this subsection provides our syntactic analyses of them, and then excludes an ellipsis-based analysis of *fe/me*-questions.

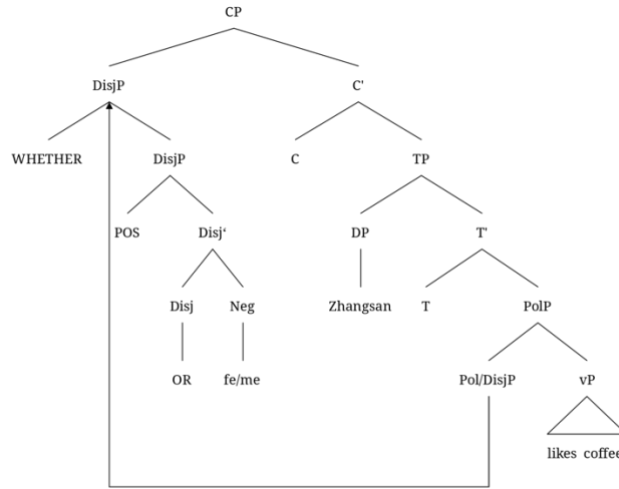
2.2.1. *Fe/me*-questions

Since *fe* and *me* in Yixing are genuine negation, we propose they are the negation in the polar DisjP parallel to *not* in English *or not*. In this polar DisjP, only the negative polarity is pronounced, but the positive polarity and ‘or’ are not (i.e., phonologically covert, leading to $\langle POS \rangle \langle OR \rangle fe/me$, where silent VIs are in $\langle \rangle$). This is not implausible because the interrogative ‘or’ in the Chinese languages can be silent when the disjuncts contrast in polarity, and the size of the disjuncts is smaller than a TP (Zhu 1985; Zhang 1990; C.-T. J. Huang 1991; C.-T. J. Huang et al. 2009).⁶ In the present analysis, the disjuncts are very small—just Polarity⁰ (Pos and Neg)—and thus satisfy these conditions.

As the polar DisjP, $\langle POS \rangle \langle OR \rangle fe/me$ is base-generated in Pol(arity)⁰, hence its sensitivity to aspect, a general property of negation in Chinese languages. This also accounts for *fe* and *me*’s incompatibility with negative propositions: both negation in the negative proposition and the polar DisjP need to occupy Pol⁰. When Pol⁰ is already occupied by negation, there is no place for the polar DisjP. While *fe* and *me* are base-generated sentence-medially in Pol⁰, we argue that they appear sentence-finally in yes/no questions because as a part of the polar DisjP, they have been pied-piped by the *wh*-movement of a covert ‘whether’:

(38)

⁶ We thank a reviewer for pointing out that the silencing of the interrogative *or* in Mandarin is restricted.



The DisjP appears sentence-finally because complementisers in Chinese generally follow their syntactic sister. There are many proposals for why this is, and they can all explain why the DisjP appears sentence-finally here. We won't go into detail, but just mention two possibilities. One is Pan's (2022) suggestion of massive fronting of the entire TP, and the other is Chomsky's (2019) suggestion that specifiers (i.e., Spec, CP here) can be merged on either side of the head because Merge in narrow syntax does not specify linear order.

We provide two pieces of evidence that *fe/me*-questions involve *wh*-movement. First, sentence-final *fe/me* can have long-distance dependencies across finite clauses and be interpreted as a yes/no question with embedded disjunctive choice:

(39) Zangsa huani Lisi huexi qie kafi fe?

Zhangsan suspect Lisi like drink coffee FE

✓Embedded reading: 'Does Zhangsan suspect that Lisi likes to drink coffee, or does Zhangsan suspect that Lisi doesn't like to drink coffee?'

✓Matrix reading: ‘Does Zhangsan suspect that Lisi likes to drink coffee, or does

Zhangsan not suspect that Lisi likes to drink coffee?’

As a reviewer pointed out, the embedded and matrix readings of (39) can be addressed with different answers. (39) can indeed be answered with ‘He suspects she doesn’t like to drink coffee’, the no-answer to the embedded reading, or ‘He doesn’t suspect that she likes to drink coffee’, the no-answer to the matrix reading.

The embedded reading is the result of *wh*-movement from the embedded Pol ((40a); silent morphemes are in $\langle \rangle$), whereas the matrix reading is the result of *wh*-movement from the matrix Pol (40b).

(40) a. Zangsa huani [CP Lisi [PolP *t_i* huexi qie kafi]] [$\langle$ WHETHER $\rangle$ $\langle$ POS $\rangle$ $\langle$ OR $\rangle$ *fe*]_i?

b. Zangsa [PolP *t_i* huani [CP Lisi huexi qie kafi]] [$\langle$ WHETHER $\rangle$ $\langle$ POS $\rangle$ $\langle$ OR $\rangle$ *fe*]_i?

Second, this long-distance dependency cannot cross an island (e.g., a relative clause), suggesting that this *wh*-movement is island-bound:

(41) **ni* pongdao-le [Relative Clause huexi qie kafi ge wosangzi] *fe*?

you meet-PERF like drink coffee GE student FE

Intended Embedded Reading: ‘Did you meet the student who likes to drink coffee, or did you meet the student who doesn’t like to drink coffee?’

Sentence (41) is ungrammatical because the covert ‘whether’ associated with the embedded polar DisjP cannot move out of the relative clause. *Fe* cannot be associated with the matrix clause because the matrix predicate is perfective.

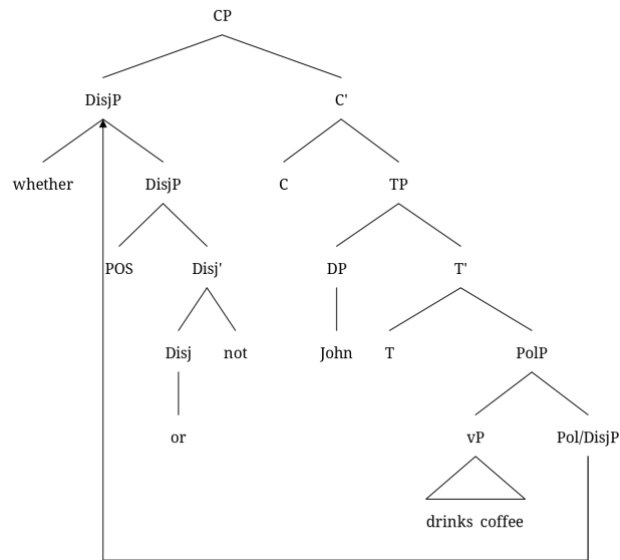
(42) **ni* pongdao-le [Relative Clause [PolP *t_i* huexi qie kafi ge wosangzi]] [$\langle$ WHETHER $\rangle$ $\langle$ POS $\rangle$ $\langle$ OR $\rangle$ *fe*]_i?

The pied-piping of the polar DisjP by the covert ‘whether’ in Yixing parallels what Wu (2022) argued to be the pied-piping of *or not* by *whether* in English. Building on Larson’s (1985) and Han and Romero’s (2004) idea that *whether* is a *wh*-word and moves, Wu further argued that *whether* pied-pipes the polar DisjP optionally:

- (43) a. I don’t know whether John likes coffee or not. *No pied-piping*
 b. I don’t know whether or not John likes coffee. *Pied-piping*

Wu assigned the following structure to (43b), which is identical to the structure we propose for Yixing *fe/me*-questions. The only difference between them is that English *whether*’s pied-piping is optional, whereas Yixing covert ‘whether’ always pied-pipes the polar DisjP.

(44)



2.2.2. *Ma*-questions

In contrast to Yixing *fe/me*-questions, Mandarin *ma*-questions do not involve any DisjP, and nothing moves there. *Ma* is base-generated in C^0 as a complementiser. Unlike *fe/me*-

questions, *ma*-questions cannot involve long-distance dependencies across finite clauses. The following sentence can only be interpreted as a matrix yes/no question, and lacks the embedded reading:

(45) Zhangsan huaiyi [CP Lisi xihuan he kafei] ma?

Zhangsan suspect Lisi like drink coffee MA

*Embedded reading: ‘Does Zhangsan suspect that Lisi likes to drink coffee or does Zhangsan suspect Lisi doesn’t like to drink coffee?’

✓Matrix reading: ‘Does Zhangsan suspect that Lisi likes to drink coffee or does Zhangsan not suspect Lisi likes to drink coffee?’

This suggests that no question particle has moved across the finite clause. Because *ma* cannot be embedded in general, it cannot appear in any island; thus, we cannot test island effects on *ma*-questions.

Before presenting the syntactic analysis of *ma*-questions, we introduce the basics of question semantics that our analysis follows. In order to capture the difference between ALTQVNS and POLQS, we argue for two different question operators, which can be considered as two different interrogative C^0 s for our purposes. The first interrogative C^0 is an identity function that has to combine with an alternative set with more than one member (46) (Kratzer and Shimoyama 2002). A question is thus an alternative set that raises an issue using pragmatic mechanisms. We assume that a disjunction denotes an alternative set with its disjuncts as the alternatives (e.g., Alonso-Ovalle 2005; Simons 2005; Beck and Kim 2007; Erlewine 2025): *coffee or tea* denotes {coffee, tea}, and *John likes coffee or not* denotes {John likes coffee, John doesn’t like coffee}. Propositions

containing a disjunction or a *wh*-constituent like *what* and *who* denote alternative sets.

This alternative set of propositions combines with the identity-function interrogative C^0 to create alternative questions (including ALTQVNs) and *wh*-constituent questions. We thus call this C^0 ALTQVN C^0 .

(46) Where $\llbracket \alpha \rrbracket = \{A, B, C \dots\}$ (undefined if a singleton set):

$$\llbracket C_{\text{ALTQVN}} \alpha \rrbracket = \{\lambda w. A(w), \lambda w. B(w), \lambda w. C(w) \dots\} \text{ (Kratzer and Shimoyama 2002:8)}$$

Besides the ALTQVN C^0 , the other interrogative C^0 (which we call POLQ C^0) has to combine with a singleton set rather than an alternative set with multiple members (47) (Hamblin 1973). It turns the content proposition that it combines with into an alternative set with two members, one being the content proposition itself, and the other being its negation. This interrogative C^0 only occurs in POLQs.

(47) Where $\llbracket \alpha \rrbracket = \{A\}$ (undefined if not a singleton set):

$$\llbracket C_{\text{POLQ}} \alpha \rrbracket = \{\lambda w. A(w), \lambda w. \neg A(w)\} \quad \text{(Hamblin 1973:50)}$$

In support of the co-existence of these two types of C^0 s, the null C^0 in *fe/me*-questions is ALTQVN C^0 , whereas *ma*, the complementiser in Mandarin *ma*-questions, is POLQ C^0 .

Evidence that *ma* is POLQ C^0 is that *ma* can only appear in yes/no questions, but not alternative questions or *wh*-constituent questions:

(48) a. Zhangsan xihuan he kafei haishi cha (*ma)?

Zhangsan like drink coffee or_{whether} tea MA?

‘Does Zhangsan like to drink coffee or tea?’

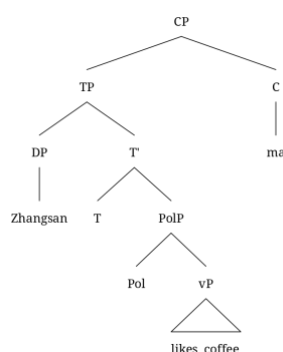
b. Zhangsan xihuan na zhong kafei (*ma)?

Zhangsan like which kind coffee MA?

‘Which kind of coffee does Zhangsan like?’

This is because *ma* has to combine with a singleton set, and propositions with a DisjP or a *wh*-constituent denote an alternative set with multiple members. The proposition in the singleton set that *ma* combines with can be positive or negative. When the proposition is negative, we get negative yes/no questions. Following is our syntactic analysis of *ma*-questions:

(49)



Section 2.1 has shown that English ALTQVNs have the identical syntactic-semantic behavior as *fe/me*-questions, whereas POLQs have the identical behavior as *ma*-questions. We highlight another parallel here: English ALTQVNs involve long-distance movement, just like *fe/me*-questions, whereas English POLQs don't, just like *ma*-questions. It has been observed that English ALTQVNs can involve long-distance dependency (Larson 1985; Han and Romero 2004). For example, in parallel to the Yixing question (39), (50) can be interpreted as a yes/no question with embedded disjunctive choice or one with matrix disjunctive choice.

(50) Did Zhangsan say that Lisi likes coffee or not?

✓Embedded reading: ‘Did Zhangsan say that Lisi likes coffee, or did Zhangsan say that Lisi doesn’t like coffee?’

✓Matrix reading: ‘Did Zhangsan say that Lisi likes coffee, or did Zhangsan not say that Lisi likes coffee?’

This ambiguity has been accounted for by different origination sites of *whether* (which is silent in matrix questions) and its subsequent *wh*-movement. The embedded reading is the result of merging *whether* in the embedded clause followed by long-distance movement (51a), whereas the matrix reading is the result of merging *whether* in the matrix clause followed by movement to Spec, CP (51b).

(51) a. <WHETHER>_i Zhangsan said [_{CP} Lisi [_{PolP} likes coffee *t_i* <POS> or not]]?

b. <WHETHER>_i Zhangsan [_{PolP} said [_{CP} Lisi likes coffee] *t_i* <POS> or not]?

In contrast to ALTQVNs, POLQs only have the matrix reading:⁷

(52) Did Zhangsan say that Lisi likes coffee?

*Embedded reading: ‘Did Zhangsan say that Lisi likes coffee, or did Zhangsan say that Lisi doesn’t like coffee?’

✓Matrix reading: ‘Did Zhangsan say that Lisi likes coffee, or did Zhangsan not say that Lisi likes coffee?’

This fact is mysterious under the common view in the literature that *whether* always originates from disjunction, but not under the current view that it doesn’t in POLQs. We

⁷ We are grateful to Maribel Romero (p.c.) for pointing out this fact to us, which we confirmed with three consultants.

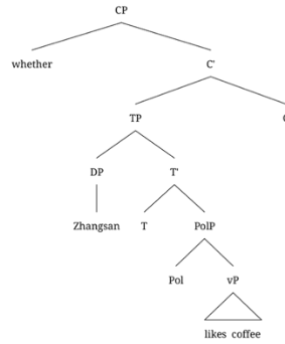
discuss two examples of the view in the literature and why this is problematic for them.

The first is Larson's (1985) idea that *or not* can be silent. This silent *or not* should be able to be merged in the matrix clause or the embedded clause, leading to the possibility of both readings like (51a-b), contrary to fact. The second is Han and Romero's (2004) idea that yes/no questions involve clausal disjunction plus ellipsis of the second disjunct and *or*. (52) should in principle be able to involve embedded clausal disjunction (53a) or matrix clausal disjunction (53b), contrary to fact.

- (53) a. <WHETHER>_i Zhangsan said [_{TP} _{t_i} Lisi likes coffee ~~or Lisi not likes coffee~~]?
 b. <WHETHER>_i [_{TP} _{t_i} Zhangsan said Lisi likes coffee ~~or Zhangsan not said Lisi likes coffee~~]?

Han and Romero (2004) raised another possibility—instead of clausal disjunction plus ellipsis, the disjunctive meaning of positive and negative propositions in these questions is provided by Karttunen's (1977b) question semantics. This is close to what we argue for. We take the contrast between the ALTQVN in (50) and the POLQ in (52) as evidence that English also has two C⁰s, ALTQVN C⁰ and POLQ C⁰. The syntactic structure of English ALTQVNs, with pied-piping, has been provided in (44). Given the absence of the embedded reading in the POLQ (52), we think English POLQs do not involve long-distance *wh*-movement, and thus assign to them largely the same syntactic analysis as *ma*-questions. Since *whether* has *wh*-morphology, we assume it is in Spec, CP, rather than POLQ C⁰ itself, which we assume is silent.

(54)



The lack of *wh*-movement in *ma*-questions and English POLQs might make us think of the lack of *wh*-movement in some genuine *wh*-in-situ constructions (e.g., English superiority-violating questions), but they are different. Genuine *wh*-in-situ constructions are subject to intervention effects by focus-sensitive operators (e.g. *even*, *only*), negation and *every* (55), but *ma*-questions and English POLQs aren't.

- (55) a. [Which student]_i didn't t_i read which book? *superiority-obeying*
 b. *[Which book] didn't which student read t_i? *superiority-violating*

As we saw in (34a-b) and (36b), negation can occur in *ma*-questions⁸ and English POLQs.

So can focus-sensitive operators like *zhi* 'only':

- (56) a. Zhangsan zhi xihuan kafei ma?
 Zhangsan only like coffee MA
 'Does Zhangsan only like coffee?'
 b. Does Zhangsan only like coffee?

⁸ As was pointed out by a reviewer, not all intervention effects and interveners are attested in Mandarin *wh*-questions (e.g., Yang 2012; Li and Law 2016). What is crucial to our point is that for all the intervention effects that do occur in Mandarin *wh*-questions and even A-not-A questions, *ma*-questions are consistently immune to them.

Therefore, *ma*-questions and English POLQs are theoretically interesting because they enrich the typology of questions. They suggest that there are at least three types of questions: (a) questions with (overt or covert) *wh*-movement (e.g., standard *wh*-constituent-questions, Yixing *fe/me*-questions, Mandarin *bu/mei*-questions, English ALTQVNs); (b) genuine *wh*-in-situ questions without any *wh*-movement (e.g., English superiority-violating questions, Japanese *wh*-in-situ; Kotek 2018); and (c) yes/no questions without any *wh*-element, and therefore no *wh*-movement or *wh*-in-situ (e.g., Mandarin *ma*-questions, English POLQs). Mandarin *ma* is base-generated in C^0 , but its derivation does not end there. In subsection 2.3 we will argue that it moves to SpeechAct^0 , a position occupied by *a* in Yixing. This will account for their unembeddability.

2.2.3. Against an ellipsis-based analysis of *fe/me*-questions

Our analysis of *fe/me*-questions assumes that *fe* and *me* are part of the polar DisjP. A plausible alternative hypothesis is that *fe/me*-questions involve underlying disjunction of positive and negative propositions by a silent ‘or_{whether}’, followed by ellipsis in the second disjunct.⁹ The sentence-final *fe/me* is then the negation in the second disjunct:

⁹ To answer a reviewer's question, the second disjunct of overt 'or_{whether}' cannot be elided at all:

(i) *Zangsa [huexi qie kafi] asi [fe ~~huexi qie~~ ~~kafi~~]? ~~asi~~

Zhangsan like drink coffee or_{whether} NEG like drink coffee

Intended: ‘Does Zhangsan like to drink coffee or not?’

Thus, the alternative ellipsis-based analysis would have to say that somehow ellipsis requires ‘orwhether’ to be silent.

(57) Zangsa [huexi qie kafi] [fe ~~huexi qie kafi~~]?
 Zhangsan like drink coffee NEG like drink coffee

‘Does Zhangsan like to drink coffee or not?’

This subsection provides three pieces of evidence against this ellipsis-based analysis. The first concerns negating the verb *you* ‘possess’, which in Yixing requires the negation *me*, not *fe*, even if the predicate is imperfective:

(58) a. Zangsa you xu.

Zhangsan possess book

‘Zhangsan has books.’

b. Zangsa {*fe/me} xu.

Zhangsan have.NEG book

‘Zhangsan does not have books.’

In an alternative question with transparent disjunction of two propositions, *me* is used in the negative proposition, not *fe*:

(59) Zangsa [you xu] [{*fe/me} xu]?
 Zhangsan have book have.NEG book

‘Does Zhangsan have books or not?’

If we apply ellipsis to the second disjunct of (59), we would expect *me* to occur sentence-finally. Contrary to this prediction, *fe* but not *me* surfaces. According to our analysis, this is because the negation in the polar DisjP is sensitive to the imperfective aspect.

(60) Zangsa you xu {fe/*me}?

Zhangsan have book FE/ME

‘Does Zhangsan have books or not?’

The second piece of evidence comes from island constraints. Alternative questions involving transparent disjunction in Yixing are not subject to subject island constraints:¹⁰

(61) [_{subject} Zangsa [be livo] asi[fe be livo]] wai niang ni kaixin?

Zhangsan give gift or NEG give gift will make you happy

‘Which will make you happy: Zhangsan giving you a gift or Zhangsan not giving you a gift?’

If *fe/me*-questions are indeed alternative questions plus ellipsis, they should not be sensitive to islands, contrary to fact (41). Also contrast (61) with (62).

(62) * [_{subject} Zangsa be livo fe] wai niang ni kaixin?

Zhangsan give gift FE will make you happy

If (62) is derived by deleting the VP in the second disjunct of (61), then (62) should be fine, contrary to fact. According to our analysis, *fe* must be pied-piped to matrix Spec, CP, and thus cannot occur in embedded clauses, as in (62). *Fe* can occur sentence-finally, as in (63), but it only has the matrix reading.

(63) [_{subject} Zangsa be livo] wai niang ni kaixin fe?

Zhangsan give gift will make you happy FE

*Embedded reading: ‘Which will make you happy: Zhangsan giving you a gift or Zhangsan not giving you a gift?’

¹⁰ Similarly, alternative questions with transparent disjunction in Mandarin Chinese are also insensitive to subject island constraints (C.-T. J. Huang et al. 2009; Holmberg 2016). For further discussion about their island (in)sensitivity, see Erlewine (2012) and R. R. Huang (2020).

✓Matrix reading: ‘Will Zhangsan’s giving you a gift make you happy or not happy?’

According to our analysis, there is a covert *wh*-operator in *fe/me*-questions. It can only move from the matrix Pol, but not from the Pol embedded in the subject because it is subject to the subject island.

Finally, the third piece of evidence comes from the adjectival marker *lao* in Yixing. In declarative sentences, a bare adjectival predicate must be marked by *lao*:

- (64) ga ge dagao haoqie *(lao).
 this CL cake tasty LAO
 ‘This cake is tasty.’

In an alternative question with disjoined adjectival predicates, *lao* cannot appear in either disjunct:

- (65) ga ge dagao[haoqie (*lao)] [fe haoqie (*lao)]?
 this CL cake tasty LAO NEG tasty LAO
 ‘Is this cake tasty or not?’

If a *fe*-question is derived by the deletion of the second disjunct of (65), then it should also reject *lao*. This prediction, however, is not borne out:

- (66) ga ge dagao haoqie lao fe?
 this CL cake tasty LAO FE
 ‘Is this cake tasty or not?’

Thus, we do not assume that ellipsis derives *fe/me*-questions but rather analyse *fe* and *me* as the negative polarities in DisjP; they appear sentence-finally because as a part of the DisjP, they are pied-piped by the null ‘whether’.

2.3. *Ma* and *a* end up in SpeechAct⁰

Section 2.1 has discussed two important differences between *fe/me*-questions and *ma*-questions. The first is that *fe/me*-questions but not *ma*-questions involve a polar DisjP, for which we have provided a syntactic analysis in section 2.2. The second difference is that *a*- and *ma*-questions cannot be embedded, but *fe/me*-questions without *a* can. This section provides an account of this fact by arguing that *ma* and *a* contribute the speech act force of requiring the addressee to answer the question (Krifka 2014; Holmberg 2016; Dayal 2025) and are in the SpeechAct (SA) head above CP.

We adopt the established view that a matrix question involves a higher SpeechAct Phrase (SAP) above CP in the left periphery (e.g., Rizzi 1997; Speas and Tenny 2003; Krifka 2014; Dayal 2025). A full root question that needs an answer from the addressee involves two semantic components (Hooper and Thompson 1973; Krifka 2014; Bhatt and Dayal 2020; Dayal 2025): a set of propositions encoded in the CP (Hamblin 1973), and a higher illocutionary operator, ASK (Dayal 2025) or QUEST (Krifka 2014), which we take to be SA⁰ in this paper. This illocutionary operator puts the addressee under the obligation to assert which proposition in the set of propositions is true.

A key property of the illocutionary operator in questions is that it typically cannot be embedded by predicates like ‘know’ and ‘find out’, but can be embedded by predicates

like ‘wonder’ and ‘ask’.^{11,12} Mandarin *ma*-questions have this property, leading us to posit that *ma*-questions have SA⁰. As we saw in (30), repeated below, *ma*-questions cannot be embedded by *zhidao* ‘know’, but it can be embedded by the predicate *wen* ‘ask’.

(30) *wo zhidao Lisi xihuan he kafei ma.

I know Lisi like drink coffee MA

Intended: ‘I know whether Lisi likes to drink coffee.’

(67) Zhangsan wen wo Lisi xihuan kafei ma.

Zhangsan ask me Lisi like coffee MA

‘Zhangsan asked me whether Lisi likes coffee.’

In parallel, Yixing *a*-questions can be embedded under ‘ask’ (68), but not under ‘know’

(28), repeated below:

(28) wo xiaoze lao Zangsa huexi qie kafi {fe/*fe a/*fa}.

I know LAO Zhangsan like drink coffee FE/FE A/FA

‘I know whether Zhangsan likes to drink coffee.’

¹¹ The reason for this distribution is beyond the scope of our paper, but see Krifka (2014) and Dayal (2025) for the further explanation.

¹² While Krifka (2014) showed that speech act questions can be embedded by predicates like *ask* and *wonder*, Dayal (2025) argued that SAP only occurs in the matrix clause, and the embedded root question has an additional level, the Perspective Phrase that is above CP but below SAP. What they agreed on is that root questions involve an XP higher than CP, making them difficult to get embedded in most cases. We follow Krifka (2014) and Holmberg (2016) in taking this higher XP as SAP. This analysis is compatible with Dayal (2025) if this XP is Perspective Phrase.

(68) Zangsa men wo Lisi huexi kafi {fe a/fa}.

Zhangsan ask me Lisi like coffee FE A/FA

‘Zhangsan asked me whether Lisi likes coffee.’

We thus take these facts to indicate that *ma*- and *a*-questions involve SAP above CP.

Apart from the embedding pattern, an additional piece of evidence that *ma*- and *a*-questions involve SAP is their incompatibility with the SFP *ne* and its Yixing counterpart *nang*. Without *a*, *nang* can occur in *fe/me*-questions.

(69) a. *Mandarin ma-question*

Zhangsan xihuan he kafei ma (*ne)?

Zhangsan like drink coffee MA NE?

‘Does Zhangsan like to drink coffee?’

b. *Yixing questions*

Zangsa huexi qie kafi {fe/*fe a/*fa/*a} nang?

Zhangsan like drink coffee FE/FE A/FA/A NANG

‘Does Zhangsan like to drink coffee?’

The SFPs *ne* and *nang* can appear sentence-finally in different types of matrix questions, including both yes/no questions and *wh*-constituent questions. While the syntactic details of these SFPs are not the focus of our paper, a common assumption is that Mandarin *ne* is not responsible for the interrogative force (Pan and Paul 2016; Paul and Pan 2017; Soh 2025): it is a head that can also take declaratives as complement, as well as *wh*-constituent questions and some yes/no questions like A-not-A (cf. Pan and Paul 2016; Paul and Pan 2017). This puts *ne* in a position above our C^0 , which we assume to be

SA⁰.^{13,14} Assuming that there can only be one SA⁰ in a clause, the incompatibility of *ma* and *a* with these SFPs suggests that *ma* and *a* are also in SA⁰, and thus they cannot all co-occur.

Since we argued in section 2.2 that *ma* is the interrogative C⁰ in POLQs, we propose that *ma* moves from POLQ C⁰ to the higher SA⁰ via head movement. In other words, the functional item *ma* is the phonological exponent of both the POLQ interrogative feature and the SA feature and surfaces in SA⁰. Here we follow the Distributed Morphology framework (DM; e.g., Halle and Marantz 1993; Marantz 2006; Embick 2010; 2015) and traditional assumptions about head movement, where functional heads can be bundled together due to head movement and realised as a single functional item.

Since *ma* always marks yes/no questions, it has long since been treated as C⁰ (cf., Lee 1986; Pan and Paul 2016; Paul and Pan 2017, following Rizzi's 1997 split CP hypothesis). While we agree that *ma* is base-generated in C⁰, our approach differs from

¹³ The analysis of *ne* and *ma* in Pan and Paul (2016) and Paul and Pan (2017) was based on a revised version of Rizzi's (1997) split CP framework, which built a three-layered CP for Chinese: Low C < Force < Attitude. *Ma* was proposed to be in the Force head, and *ne* here is the Attitude head (Pan and Paul 2016; Paul and Pan 2017). In our analysis, there is only one C⁰, and our C⁰ in questions roughly corresponds to the Force head in Pan and Paul (2016) and Paul and Pan (2017); the head above this interrogative C⁰ is called SA⁰ in our analysis. The point we take from these studies is that *ne* is situated in a functional head above the head for the interrogative force (i.e., the head with the function of clause typing).

¹⁴ There are two types of *ne* in Mandarin, and here we only focus on the *ne* that can occur in a question. For a formal analysis of the two *nes* in Mandarin (as a low C and an Attitude head, respectively), see Pan and Paul (2016) and Paul and Pan (2017).

the previous studies in the following two ways. First, we argue that *ma* is C⁰ exclusively associated with POLQs, to account for the lack of DisjP in *ma*-questions and the impossibility of *ma* in alternative questions and *wh*-constituent questions, a fact that the previous approaches that considered *ma* to be generic interrogative C⁰ fail to account for. Second, we argue that *ma* moves from C⁰ to SA⁰, to explain why *ma* can only appear in root questions, and why it cannot occur with the particle *ne* that is in a head position higher than C⁰.¹⁵

We now turn to our analysis of the SFP *a* in Yixing. The fact that neither Mandarin *ma* nor Yixing *a* can be embedded or co-occur with other SA particles seems to suggest that Yixing *a* is also in SA⁰. We argue that *a* is base-generated in SA⁰, unlike *ma*, which moves to SA⁰ from C⁰. Our argument is based on a key difference between *ma* and *a*: unlike *ma*, which cannot occur in *wh*-constituent questions or alternative questions, *a* can:

- (70) a. ni huexi qie die a?
 you like eat what A
 ‘What do you like eating?’

¹⁵ A reviewer asked whether *ma*-questions should be analysed as tag questions, noting that neither type of question can be embedded or answered by ‘It is so.’ Three observations suggest that *ma*-questions are not tag questions. First, there is a clear prosodic break before the tag (i.e., in Mandarin, *shi ma?* ‘is it so?’ or *dui ma?* ‘right?’), but not before *ma* in *ma*-questions. Second, *ma*-questions, but not tag questions, can indicate the speaker’s incredulity or surprise. Yuan and Hara (2019) highlighted this contrast using the incredulity-marking adverb *nandao* ‘difficult-say’: it can occur in *ma*-questions, but not in tag questions. Finally, Yuan and Hara also showed that the adverb *bijing* ‘after all’ can appear in tag questions but not in *ma*-questions.

b. ni huexi qie kafi asi cuo a?

you like eat coffee or tea A

‘Do you like to drink coffee or tea?’

This is because *a* is in SA⁰ and selects an interrogative CP, regardless of whether that CP is an ALTQVN or a POLQ. In other words, *a* does not care whether the head of its complement is ALTQVN C⁰ or POLQ C⁰. Thus, as predicted, *a* is not only compatible with ALTQVNs and *wh*-constituent questions, but also POLQs (i.e., yes/no questions without *fe* and *me*). We can use the diagnostics in section 2.1 to show that these yes/no questions without *fe* and *me* but with *a* are indeed POLQs. First, they can be answered by ‘It is so’:

(71) A: Zangsa huexi qie kafi de a?

Zhangsan like drink coffee DE A

‘Does Zhangsan like to drink coffee?’

B: si ge.

COP GE

‘It is so.’

Second, these questions can take a negative proposition:

(72) Zangsa fe huexi qie kafi a?

Zhangsan NEG like drink coffee A

‘Does Zhangsan not like to drink coffee?’

Our other diagnostics, the unconditional test and the ‘doubt’-test cannot apply to these questions because *a* cannot be embedded in general, as we saw in (28).

2.4. Mandarin *bu/mei*-questions

Yixing *fe/me*-questions may remind Mandarin speakers of what we call *bu/mei*-questions in Mandarin (e.g., Zhang 1990; Cheng et al. 1996; C.-T. J. Huang et al. 2009), which insert negation at the end of a declarative. These questions were also called VP-NEG questions by C.-T. J. Huang et al. (2009). This subsection will show that *bu/mei*-questions behave like Yixing *fe/me*-questions in some aspects but like Mandarin *ma*-questions in others. This leads to our proposal for *bu/mei*-questions that is essentially a hybrid of our analyses of *fe/me*- and *ma*-questions.

We begin by discussing two similarities between Yixing *fe/me*-questions and *bu/mei*-questions. Like *fe/me*-questions, Mandarin *bu/mei*-questions end in an SFP that appears to be identical with the negations, *bu* and *mei* in the language. Like their counterparts in Yixing, negation *mei* and the corresponding SFP *mei* are used with perfective aspect, whereas negation *bu* and its corresponding SFP are used elsewhere.

(73) a. Zhangsan xihuan he kafei bu?

Zhangsan like drink coffee BU

‘Does Zhangsan like to drink coffee?’

b. Zhangsan jintian he-le kafei mei?

Zhangsan today drink-PERF coffee MEI

‘Did Zhangsan drink coffee today?’

Besides the insertion of negation at the end of a declarative, Mandarin *bu/mei*-questions share another similarity with Yixing *fe/me*-questions: they involve a polar DisjP, as can

be shown with our diagnostics. As is observed by C.-T. J. Huang et al. (2009), *bu/mei*-questions cannot accept short answers like *shi de* ('It is so'):

(74) A: Zhangsan he-le kafei mei?

Zhangsan drink-PERF coffee MEI

Did Zhangsan drink coffee?

B: #Shi de.

COP DE

'It is so.'

Bu/mei-questions cannot take a negative proposition, again suggesting that they involve a polar DisjP (Cheng et al. 1996):

(75) *Zhangsan bu xihuan he kafei bu?

Zhangsan NEG like drink coffee BU

Intended: 'Does Zhangsan not like to drink coffee?'

Given these similarities between *bu/mei*-questions and *fe/me*-questions, it may be tempting to assume that *bu/mei*-questions should have the same syntactic analysis as Yixing *fe/me*-questions. However, *bu*-questions behave differently from *fe/me*-questions and similarly to Mandarin *ma*-questions in other aspects. The first is that unlike Yixing *fe/me*-questions but like Mandarin *ma*-questions, *bu*-questions cannot be embedded (C.-T. J. Huang et al. 2009); in contrast, *mei*-questions can be.¹⁶

(76) a. *wo zhidao [ta xihuan kafei bu].

I know he like coffee BU

¹⁶ We are grateful to a reviewer for pointing this out.

Intended reading: ‘I know whether he likes coffee.’

b. wo zhidao [ta mai-le kafei mei].

I know he buy-PERF coffee MEI

‘I know whether he has bought coffee.’

Second, recall from section 2.3 that Mandarin *ma*-questions and Yixing *a*-questions cannot be followed by the SA SFP *ne* or *nang* (69). Neither can *bu*-questions; in contrast, *mei*-questions can:

(77) a. *Zhangsan xihuan he kafei bu ne?

Zhangsan like drink coffee BU NE

‘Does Zhangsan like to drink coffee?’

b. Zhangsan he-le kafei mei ne?

Zhangsan drink-PERF coffee MEI NE

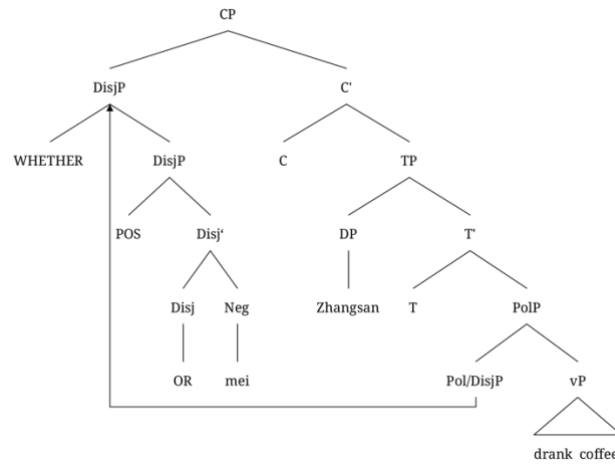
‘Did Zhangsan drink coffee?’

Our account of the incompatibility of *ma* and *a* with these SA SFPs was that *ma* and *a* are also in SA⁰, and there can only be one SA⁰ in a sentence. Extending this account to Mandarin *bu/mei*, *bu* is in SA⁰, but not *mei*.

Thus, on the one hand, *bu/mei*-questions pattern like Yixing *fe/me*-questions but not *ma*-questions in involving a polar DisjP. On the other hand, *bu*-questions pattern like *ma*-questions but not *fe/me*-questions in being unembeddable and incompatible with the SFP *ne*. We take the former type of evidence as indication that *bu* and *mei* are the negative polarity in the polar DisjP, just like *fe* and *me*. We take the latter type of evidence as

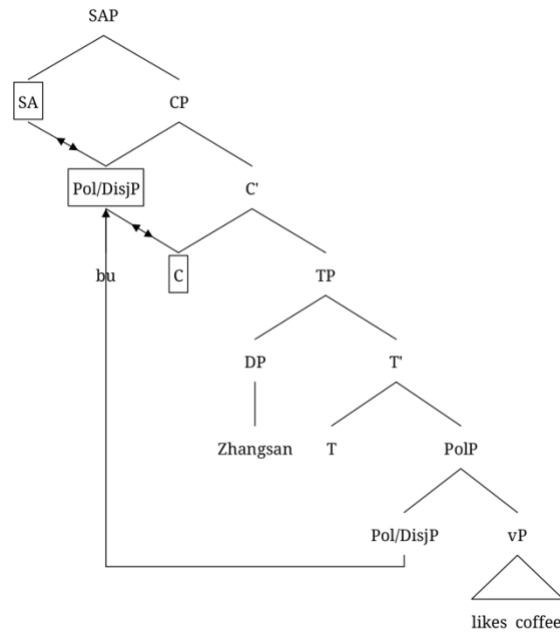
indication that *bu* has SA feature, but not *mei*. Thus, we assign the same analysis to *mei*-questions as to *fe/me*-questions:

(78)



Assuming that only heads can undergo head movement, we thus think that in *bu*-questions, the polar DisjP in Spec, CP and C⁰ fuse into a single head, and this head then fuses with SA⁰ (79), resulting in the form of a single head.

(79)



Our analysis of *bu/mei*-questions involves base-generation of *bu/mei* in the TP domain and its movement to a higher position. This is also the assumption of Cheng et al. (1997) and C.-T. J. Huang et al. (2009). There are two important differences between our analysis and this literature, however. First, while Cheng et al. did assume that negation is base generated in the TP domain, they did not explain why. For them, the negation is not part of a polar DisjP, but marks the negative polarity of a declarative sentence, but it is not clear why the yes/no question should be derived from a negative sentence, or how the negative sentence leads to the yes/no question meaning. In our analysis, the negation is base-generated in the TP because it is part of the DisjP. The other difference is that Cheng et al. took C^0 to be the final landing site of negation, whereas we think the final landing site of *bu* is the higher SA^0 . Since C^0 should be allowed in embedded clauses, Cheng et al. could not explain why *bu*-questions are unembeddable.

In sum, this section has argued that Yixing *fe/me*-questions involve a polar DisjP, whereas Mandarin *ma*-questions don't involve any DisjP or *wh*-movement. We then argued that *ma* moves from C⁰ to SA⁰. Combining the properties of both *fe/me*- and *ma*-questions, *bu*-questions involve both polar DisjP and fusion of C⁰ and SA⁰.

Our analysis provides morphological evidence that there are two different interrogative C⁰s. POLQ C⁰ is *ma* in Mandarin, whereas ALTQVN C⁰ is null in Yixing. The fact that C⁰ is null in A-not-A and *shifou*-questions (to be discussed in section 3) and *wh*-constituent questions in Mandarin suggests that ALTQVN C⁰ is also null in Mandarin. The two interrogative C⁰s also have different syntactic properties: only ALTQVN C⁰ requires *wh*-movement; POLQ C⁰ doesn't.

3. Whether the DisjP undergoes overt or covert *wh*-movement

Having focused on Mandarin *ma*-questions and Yixing *fe/me*-questions so far, this section examines three other yes/no question constructions across the Chinese varieties—A-not-A questions, *shifou*-questions and *a/ke*-questions. We will first apply our diagnostics of polar DisjP to these three question types, and show that they are ALTQVNs and involve a DisjP. But these three question types have a key difference from Yixing *fe/me*-questions and Mandarin *bu/mei*-questions, the other ALTQVNs that we have examined, in that the polar DisjP in A-not-A questions, *shifou*-questions and *a/ke*-questions appears clause-medially rather than clause-peripherally. We show that despite appearing sentence-medially, this polar DisjP is sensitive to islands, leading us to posit the second parametric variation between *fe/me*-questions on the one hand and A-not-A, *shifou*-questions and *a/ke*-questions on the other: 'whether'-movement is overt in the

former but covert in the latter. This is important because we show that both overt and covert *wh*-movements are attested as the default question strategy in the Chinese language family, contrary to previous assumption. The (c)overtness of ‘whether’-movement is a parameter across the Chinese varieties, in the same way that the (c)overtness of *wh*-constituent-movement is between English and Mandarin Chinese.

We begin by introducing the background on A-not-A, *shifou*- and *a/ke*-questions. Mandarin A-not-A questions are created by reduplicating the first syllable of the verb and inserting negation immediately after the reduplicant. As in *bu/mei*-questions, the negation inserted in A-not-A questions is sensitive to the aspect of the sentence, where *mei* is used with the perfective aspect, and *bu* is the negation elsewhere:

(80) a. Zhangsan xi bu xihuan he kafei?

Zhangsan RED NEG like drink coffee

‘Does Zhangsan like to drink coffee?’

b. Zhangsan he mei he kafei?

Zhangsan RED NEG drink coffee

‘Did Zhangsan drink coffee?’

As discussed with *bu/mei*-questions in section 2.4, negation in declaratives in Mandarin is sensitive to aspect. This leads us to think that *bu* and *mei* in A-not-A questions are genuine negations. Because multisyllabic words in Mandarin can sometimes be further divided into meaningful monosyllabic morphemes, some may not consider A-not-A questions to involve phonological reduplication, but rather copying of a morpheme that happens to be monosyllabic. Following C.-T. J. Huang (1991), C.-T. J. Huang et al.

(2009) and others, we believe A-not-A questions do involve genuine reduplication because it can reduplicate the first syllable of disyllabic words that cannot be further subdivided into morphemes (81a), the first syllable of loanwords like *PUA* that crucially cannot be further subdivided (*PUA* was adapted from English *pick-up artist*, and now means ‘abuse mentally, manipulate’ in colloquial Mandarin) (81b),¹⁷ and even in code-switching contexts (81c), where speakers’ reduplication patterns have been shown to respect sonority effects (Liu 2019).

- (81) a. Zhangsan ma bu mahu?
 Zhangsan RED NEG careless
 ‘Is Zhangsan careless?’
- b. Zhangsan pi bu piyu’ei ta?
 Zhangsan RED NEG PUA her
 ‘Does Zhangsan manipulate her?’
- c. zhe ge cai fre bu fresh?
 this CL vegetable RED NEG fresh
 ‘Is this vegetable fresh?’

In addition to the first syllable of the verb, the entire verb or the VP can also be reduplicated (C.-T. J. Huang 1991). This paper focuses on cases where only the first

¹⁷ (81b) is a naturally occurring example the authors heard on a reality TV show.

syllable is reduplicated, and we assume that the other cases may also involve an ALTQVN C⁰ and possibly a DisjP (e.g., V-, VP-, or TP-disjunction).¹⁸

Besides the A-not-A strategy, Mandarin can insert *shifou* before the predicate to create a yes/no question.

(82) a. Zhangsan shi fou xihuan he kafei?

Zhangsan COP NEG like drink coffee

‘Does Zhangsan like to drink coffee?’

Shifou can be morphologically decomposed into the copula and focus marker *shi*, and *fou*, negation in Ancient Chinese. *Shifou*-questions in Contemporary Mandarin therefore have a very formal connotation and are not part of casual speech.

There is another type of yes/no questions, what we call *a/ke*-questions, that have been attested in a variety of Chinese varieties, including varieties of Wu in Suzhou (Chao 1928), Taicang and Changshu, etc., Southwestern Mandarin in Kunming, Lower Yangtze Mandarin (also called Jianghuai Mandarin) in Nanjing and Hefei (e.g., Zhu 1985; Zhang 1990; C.-T. J. Huang 1991). In Suzhou, Taicang, Changshu and Nanjing, the morpheme *a* is inserted before the verb of a declarative sentence to create a yes/no question (7),

¹⁸ It is possible that syntactic and postsyntactic processes can apply in this DisjP, such as ellipsis (C.-T. J. Huang 1991; C.-T. J. Huang et al. 2009) and Right Node Raising (Guo 2025).

repeated below. In Kunming and Hefei, the morpheme *ke* is inserted before the verb

(83).^{19, 20}

(7) Zhangsan a xihuan huo kafei? *Nanjing*

Zhangsan A like drink coffee

‘Does Zhangsan like to drink coffee?’

(83) Zhangsan ke suong he kafei? *Hefei*

Zhangsan KE like drink coffee

‘Does Zhangsan like to drink coffee?’

Recall from section 2 that we used four diagnostics of DisjP to contrast Mandarin *ma*-questions with Yixing *fe/me*-questions. We will now apply those diagnostics to A-not-A, *shifou*- and *a/ke*-questions, and show that they involve a polar DisjP and are ALTQVNs.²¹

¹⁹ We mark the particles as *ke* and *a*, but their actual phonetic realisation varies slightly across varieties. For current purposes, we do not go into further phonetic detail here.

²⁰ Following a reviewer’s question, we would like to clarify that none of our 17 consultants of Standard Mandarin (including *Putonghua* and *Guoyu*; ranging from diverse geographic regions across China) uses *a/ke*-questions productively, and thus we assume that Standard Mandarin does not have *a/ke*-questions, following a long tradition in both descriptive and generative works (e.g., Zhu 1985; Zhang 1990; C.-T. J. Huang 1991; Zhu 1991; C.-T. J. Huang et al. 2009). Standard Mandarin does have frozen, idiomatic expressions that look like *ke*-questions such as *Ke bu shi ma?* ‘Exactly.’ but we think this may be due to historical layering or literary borrowing by written vernacular Chinese from varieties that do have *ke*-questions.

²¹ Not all yes/no questions with the surface form of a clause-medial, pre-predicate question particle in Chinese varieties are ALTQVNs like *a/ke*-questions. For example, Southern Min can insert *kam* in the position of *a/ke*.

First, recall that POLQs but not ALTQVNs can be answered by ‘so’ because ‘so’ requires the question to highlight only one proposition. Zhu (1985) observed that neither A-not-A questions like (80) nor *a/ke*-questions like (7) and (83) can be answered by *shi de* ‘It is so’ or their equivalents in the varieties; instead, the verb must be repeated in the polar answer. Below are more data from Nanjing:

(84) A: li a xihuan huó kafei (a)?

you A like drink coffee A

‘Do you like to drink coffee?’

B1: #shi dei.

COP DEI

‘It is so.’

B2: xihuan.

like

‘Yes.’

In contrast, *shifou*-questions can be answered by *shi (de)*, but we think this may be analysed as an answer that repeats the verb, which happens to be the copula *shi*:

But as a reviewer noted, *kam*-questions differ from *a/ke*-questions in that they can be negative. Other differences between *a/ke*-questions and *kam*-questions include: *kam*-questions but not *a/ke*-questions can be biased questions that convey the speaker’s incredulity; *kam* but not *a/ke* can co-occur with DisjP; *kam* but not *a/ke* can appear sentence-initially and before high adverbials like ‘today’ (Lau 2010). While it is beyond the scope of this paper to provide a detailed analysis of *kam*-questions, the diagnostics proposed in our study may be used in the future to determine whether a yes/no question is an ALTQVN.

(85) A: ni shi fou xihuan he kafei?

you COP NEG like drink coffee

‘Do you like to drink coffee?’

B: shi (de).

COP DE

‘Yes.’

Thus, answerability by ‘so’ suggests that A-not-A questions and *a/ke*-questions are ALTQVNS, but does not tell us what type *shifou*-questions are.

The second diagnostic we used to differentiate ALTQVNS and POLQs was embeddability by ‘doubt’, which requires its complement to highlight a single proposition. A-not-A, *shifou*- and *a/ke*-questions can be embedded under verbs like ‘know’ (86), but not under ‘doubt’ (87).²²

²² Y. Huang (2020) provided the opposite judgment, and claimed that ‘doubt’ can embed A-not-A questions:

(i) xiaohuanghuaiyi xiaotang shi bu shi xihuan xiaoqiang.

Huang suspect Tang RED NEG COP like Qiang

‘Huang suspects whether Tang likes Qiang.’

(Based on Y. Huang 2020:6)

We think there is a confound here: *huaiyi* is ambiguous between ‘doubt’ and ‘be uncertain about’. The latter interpretation does not require its complement to highlight only one proposition, and can thus embed A-not-A questions. *Huaiyi* in (i) should be interpreted as ‘be uncertain about.’ To tease these two readings apart, we can build a context where ‘be uncertain about’ is not possible because it requires a large degree of uncertainty, such as the following, and here A-not-A and *shifou*-questions are not possible:

(ii) Context: The police have found strong evidence that Zhangsan stole the book, and thus has reasonable suspicion that he did, though they cannot be fully certain.

- (86) a. Lisi bu zhidao Zhangsan xi bu xihuan he kafei.
 Lisi NEG know Zhangsan RED NEG like drink coffee
 ‘Lisi doesn’t know whether Zhangsan likes to drink coffee.’
- b. Lisi bu zhidao Zhangsan shi fou xihuan he kafei.
 Lisi NEG know Zhangsan COP NEG like drink coffee
 ‘Lisi doesn’t know whether Zhangsan likes to drink coffee.’
- c. Lisi bu xiaodei Zhangsan a xihuan huo kafei.
 Lisi NEG know Zhangsan A like drink coffee
 ‘Lisi doesn’t know whether Zhangsan likes to drink coffee.’
- d. Lisi bu xiaode Zhangsan ke suong he kafei.
 Lisi NEG know Zhangsan KE like drink coffee
 ‘Lisi doesn’t know whether Zhangsan likes to drink coffee.’
- (87) a. #Lisi huaiyi Zhangsan xi bu xihuan he kafei.
 Lisi suspect Zhangsan RED NEG like drink coffee
 Intended: ‘Lisi suspects whether Zhangsan likes to drink coffee.’

-
- a. #jingcha huaiyi Zhangsan tou mei tou shu.
 police suspect Zhangsan RED NEG steal book
 Intended: ‘The police suspects that Zhangsan stole the book.’
- b. #jingcha huaiyi Zhangsan shi fou tou-le shu.
 police suspect Zhangsan COP NEG steal-PERF book
 Intended: ‘The police suspects that Zhangsan stole the book.’

b. #Lisi huaiyi Zhangsan shi fou xihuan he kafei.

Lisi suspect Zhangsan COP NEG like drink coffee

Intended: 'Lisi suspects whether Zhangsan likes to drink coffee.'

b. #Lisi huaiyi Zhangsan a xihuan huo kafei.

Lisi suspect Zhangsan A like drink coffee

Intended: 'Lisi suspects whether Zhangsan likes to drink coffee.'

c. #Lisi huaiyi Zhangsan ke suonghe kafei.

Lisi suspect Zhangsan KE like drink coffee

Intended: 'Lisi suspects whether Zhangsan likes to drink coffee.'

Third, given that ALTQVNs can be used as unconditionals, but POLQs cannot, the possibility of A-not-A, *shifou*- and *a/ke*-questions in unconditionals suggests that they are ALTQVNs:

(88) a. buguan ni mingtian lai bu lai wo jia chi fan,

no.matter you tomorrow RED NEG come my home have meal

wo jintian dou yao mai henduo cai de.

I today DOU will buy many groceries DE

'Whether you're coming for a meal, I'll buy a lot of groceries today anyway.'

b. buguan ni mingtian shi fou lai wo jia chi fan,

no.matter you tomorrow COP NEG come my home have meal

wo jintian dou yao mai henduo cai de.

I today DOU will buy many groceries DE

'Whether you're coming for a meal, I'll buy a lot of groceries today anyway.'

c. buguan li mingtian a lai wo jia chi fan,
 no.matter you tomorrow A come my home have meal
 wo jintian dou yao mai haoduo cai.

I today DOU will buy many groceries

‘Whether you’re coming for a meal, I’ll buy a lot of groceries today anyway.’

d. buguan ni mingtian ke lai wo jia chi fan,
 no.matter you tomorrow KE come my home have meal
 wo jintian dou hui mai haoduo cai.

I today DOU will buy many groceries

‘Whether you’re coming for a meal, I’ll buy a lot of groceries today anyway.’

Finally, recall that ALTQVNS cannot ask about a negative proposition because in these questions, Pol⁰ has already been occupied by the polar DisjP. A-not-A and *a/ke*-questions require a positive proposition, again suggesting that they are ALTQVNS:

(89) a. *Zhangsan xi bu bu xihuan he kafei?

Zhangsan RED NEG NEG like drink coffee

Intended: ‘Does Zhangsan not like to drink coffee?’

b. *Zhangsan a bu xihuan he kafei?

Zhangsan A NEG like drink coffee

Intended: ‘Does Zhangsan not like to drink coffee?’

c. *Zhangsan ke bu suong he kafei?

Zhangsan KE NEG like drink coffee

Intended: ‘Does Zhangsan not like to drink coffee?’

Shifou-questions can take a negative proposition, but we don't think this necessarily suggests that *shifou*-questions are POLQs.

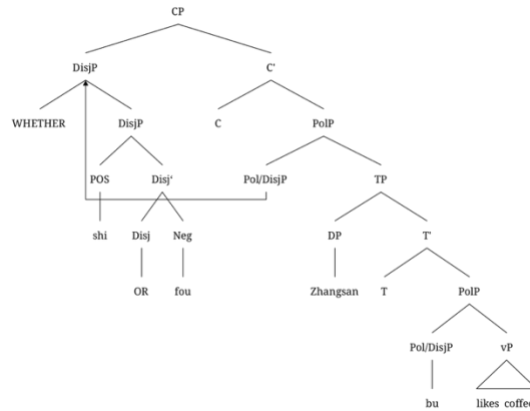
(90) Zhangsan shi fou bu xihuan he kafei?

Zhangsan COP NEG NEG like drink coffee

'Does Zhangsan not like to drink coffee?'

Following Schaffar and Chen (2001), Ye (2021), Tsai and Yang (2026) and others, there are two Pol⁰s in Mandarin, one between T and v as we have assumed, and the other above T. *Shi-bu-shi* 'COP-NEG-COP', the A-not-A form of the copula *shi*, has been argued to be located in the high PolP. We consider *shifou* to be very similar to *shi-bu-shi*, and thus assume *shifou* to also be in the high PolP. Therefore, it can scope over a negative proposition, and the low Pol⁰ is negative (91). (91) represents *shifou*'s movement to interrogative CP, a step we will argue for shortly.

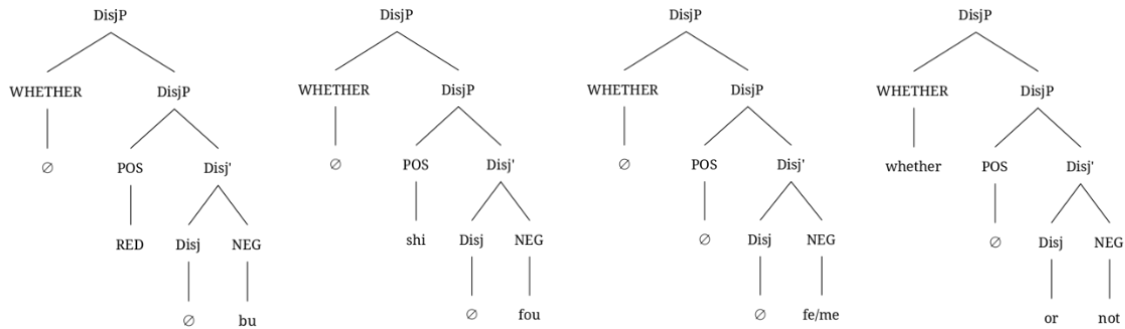
(91)



The final test is thus not conclusive for *shifou*-questions, and does not necessarily indicate they are ALTQVNs or POLQs. The four tests, taken together, have suggested that A-not-A, *shifou*- and *a/ke*-questions involve a polar DisjP. We think that the A-not part of

A-not-A, *shifou*, *a* and *ke* are in fact the polar DisjP themselves. Section 4 will expand our view on *a* and *ke*, and here we present our view on the internal syntax of A-not-A and *shifou*. We think that *A-not* should be analysed as disjunction by a silent ‘or’ of the positive polarity, a reduplicant, and the negative polarity; a silent ‘whether’ is in Spec, DisjP (92), and *shifou* should be analysed as disjunction by a silent ‘or’ of the positive polarity, *shi*, and the negative polarity, *fou* (93). This parallels the structure of Yixing *fe/me* (94), whose positive polarity is also silent, and English *whether or not* (95), where only the positive polarity is silent.

- (92) *Structure of A-not* (93) *Structure of shi-fou* (94) *Structure of fe/me* (95) *Structure of or not*



Since the negative polarity, *fou*, in *shifou* is not independently used as predicate negation in Contemporary Mandarin, it does not change based on the event aspect as *bu* and *mei* do. Thus, we cannot use the same type of evidence as in *fe/me*- and A-not-A questions to argue that *fou* is a Vocabulary Item on its own, and that *shi-fou* consists of multiple VIs. But we observe that *fou* can merge with modals such as *neng* ‘can’ and *hui* ‘will’ to generate *nengfou* ‘can or not’ and *huifou* ‘will or not’, suggesting that *fou* realises the

negative polarity in the polar DisjP and may combine with other VIs in the positive polarity.

As for A-not-A, since the negation there varies based on event aspect, we think that negation is a VI that realises the negative polarity. We argue that the positive polarity reduplicates the verb to support the negative polarity, a prosodically weak particle.²³ This parallels *do*-support in English, but rather than inserting a dummy word *do*, Mandarin reduplicates the first syllable of the verb. Evidence for our analysis comes from other constructions across the Chinese varieties: in general, the Chinese varieties use reduplication as a strategy to support a prosodically weak particle. We see this strategy deployed not only in yes/no questions, but also in other constructions, leading to the A-Particle-A order. For example, in contrastive sentences in Shanghainese and Shanghai Mandarin, the first syllable of the predicate under contrast is reduplicated, after which the prosodically light and toneless particle *me* is inserted (96).

(96) a. Zhangsan xi me xihuan de, tiao me tiaoti de.

Zhangsan RED ME like DE RED ME picky DE

‘Zhangsan does like her, but is still picky with her.’

b. Zhangsan ma me mahu de, shiqing zuo me daoshi zuowan de.

²³ Another possible analysis is verb doubling, where the verb moves, and multiple copies of the verb need to be pronounced, e.g. to support a bound morpheme (e.g. van Riemsdijk 1989; Abels 2000; Landau 2006) or due to morphological fusion (Nunes 2004; Cheng 2007; Corver and Nunes 2007). We do not pursue this analysis because there is no evidence that the verb has moved to the positive polarity (i.e. the first disjunct of the polar DisjP).

Zhangsan RED ME careless DE thing RED ME still finish DE

‘Zhangsan is careless, but he does finish his work.’

Likewise, in concessive sentences in Mandarin, the first syllable of the predicate is reduplicated, after which the focus marker *shi* is inserted (97).²⁴ We therefore think that the reduplication in A-not-A questions is an instantiation of the general strategy that the Chinese languages use to support a prosodically weak morpheme.

(97) a. zhe kafei, hao shi haohe, dan gui ye shi zhen gui.

this coffee RED SHI delicious but RED also SHI really expensive

‘This coffee does taste good, but it is also very expensive.’

b. Zhangsan ma shi mahu-le dian, danshi ta mali ya.

Zhangsan RED SHI careless-LE a.little but he quick YA

‘Zhangsan is careless, but he is quick.’

In this analysis, reduplication is to support the phonologically weak negation morpheme. Following this analysis, we can imagine a logically possible extreme situation: the negation is so weak that it is null, and requires support by the reduplicated syllable in the positive polarity. If that happens, then the A-not-A form will become A-A. This form

²⁴ These concessive sentences are not to be confused with what Cheng and Vicente (2013) called verb doubling clefts, where a verb is clefted and both its copies are pronounced. In verb doubling clefts, the subject may intervene between the copula *shi* and the low copy of the verb, but this is not possible in concessive sentences:

(i) *Ma shi Zhangsan mahu.

RED SHI Zhangsan careless

Intended: ‘As for being careless, Zhangsan is indeed careless.’

indeed exists in many varieties (e.g., Xiantao, Xiaogan, and Anlu of Hubei Province, Wang 2025; Shaoxing, Sheng County and Suizhou, Zhu 1991):

- (98) a. ni xi xihuan wo? *Xiantao*
 you RED like I
 ‘Do you like me?’
- b. ta piao piaoliang? *Xiaogan*
 she RED pretty
 ‘Is she pretty?’ (Wang 2025:540)

If the negation is indeed supported by reduplication, we may expect this negation to attach to its prosodic host (i.e., the reduplicated syllable) through fusion. We see phonological evidence of this fusion. According to Zhu (1991), Hakka Chinese spoken in Liancheng, Changting and Yudu can also use both A-not-A and A-A forms. Zhu showed that in the A-A form, the negation has been fused into the reduplicated syllable: while the segmental form of negation is lost, its tone is retained and appears on the reduplicated syllable. Specifically, the tone of the reduplicated syllable (i.e., the prosodic host) becomes Tone 35, the tone of the negation word, regardless of this syllable’s underlying tone. Fusion of negation onto the reduplicated syllable in A-not-A questions is also attested in other varieties like Fuzhou (Zhu 1991).

Our analysis of A-not-A contrasts with C.-T. J. Huang (1991), C.-T. J. Huang et al. (2009) and Holmberg (2016), who saw the A-not-A morpheme together as a head with unvalued [Polarity]-feature, and this head just happens to look like negation. Our tests involving answerability by ‘so’ and embeddability under unconditionals and ‘doubt’ have

suggested that A-not-A questions involve two propositions, one positive and one negative, not just one proposition with unvalued [Polarity]-feature as proposed in those previous studies. According to our analysis, it is not accidental that there is a negation-looking morpheme in A-not-A, *shifou*- and Yixing *fe/me*-questions, but rather, this is the negative polarity that is disjoined with the positive polarity in the polar DisjP. This accounts for their sensitivity to aspect, just like negation in declaratives.

Paul (2026) also pointed out that the negation marker in A-not-A questions cannot be inserted as a result of a reduplication rule at PF. She argued that the negation marker is instead merged in narrow syntax, as part of the ‘negated verbal projection,’ which is later merged with the Q head in the sense of C.-T. J. Huang (1982) and C.-T. J. Huang et al. (2009). Although Paul’s approach avoids the problem of treating the negation marker as an element inserted at PF, it lacks independent motivation for positing a negated VP in the derivation of a yes/no question, and the presence of a negated verbal projection in this analysis would result in a negative question, contrary to the fact that A-not-A questions are neutral questions.

Having shown that A-not-A, *shifou*- and *a/ke*-questions are ALTQVNs just like Yixing *fe/me*-questions, we note an obvious difference between them: the A-not-A morpheme, *shifou*, *a* and *ke* appear sentence-medially, whereas *fe* and *me* appear sentence-finally, which, as we have argued in section 2, is due to overt *wh*-movement. Following C.-T. J. Huang (1991), C.-T. J. Huang et al. (2009) and Holmberg (2016), we think that the A-not-A morpheme, *shifou*, *a* and *ke* also *wh*-move, but covertly. When embedded, these morphemes can lead to matrix questions, suggesting that they move long-distance from

the embedded clause to the matrix Spec, CP. These questions can be answered by ‘He claims that he does’, the yes-answer, or ‘He claims that he doesn’t.’, the no-answer.

(99) a. Zhangsan xuancheng ta xi bu xihuan he kafei?

Zhangsan claim he RED NEG like drink coffee

‘Does Zhangsan claim he likes to drink coffee or does he claim he doesn’t?’

b. Zhangsan xuancheng ta shi fou xihuan he kafei?

Zhangsan claim he COP NEG like drink coffee

‘Does Zhangsan claim he likes to drink coffee or does he claim he doesn’t?’

c. Zhangsan xuancheng ta a xihuan huo kafei?

Zhangsan claim he A like drink coffee

‘Does Zhangsan claim he likes to drink coffee or does he claim he doesn’t?’

d. Zhangsan xuancheng ta ke suong he kafei?

Zhangsan claim he KE like drink coffee

‘Does Zhangsan claim he likes to drink coffee or does he claim he doesn’t?’

But when these morphemes are embedded in an island, the sentence becomes ungrammatical, suggesting that their movement is island-bound:

(100) a. *ta renshi-le [relative clause xi bu xihuan he kafei de xuesheng]?

he meet-PERF RED NEG like drink coffee DE student

Intended reading: ‘Did he meet a student who likes coffee, or did he meet a student who doesn’t like coffee?’

b. *ta renshi-le [relative clause shi fou xihuan he kafei de xuesheng]?

he meet-PERF COP NEG like drink coffee DE student

c. *ta renshi-le [relative clause a xihuan huo kafei de xuesheng]?

he meet-PERF A like drink coffee DE student

d. *ta renshi-le [relative clause ke suong he kafei de xuesheng]?

he meet-PERF KE like drink coffee DE student

(101) a. *[subject Zhangsan xi bu xihuan he kafei] bijiao hao (ne)?

Zhangsan RED NEG like drink coffee more good NE

Intended reading: ‘Is it better that Zhangsan likes to drink coffee, or better that he doesn’t like to drink coffee’

b. *[subject Zhangsan shi fou xihuan he kafei] bijiao hao (ne)?

Zhangsan COP NEG like drink coffee more good NE

c. *[subject Zhangsan a xihuan huo kafei] bijiao hao (ne)?

Zhangsan A like drink coffee more good NE

d. *[subject Zhangsan ke suong he kafei] bijiao hao (ne)?

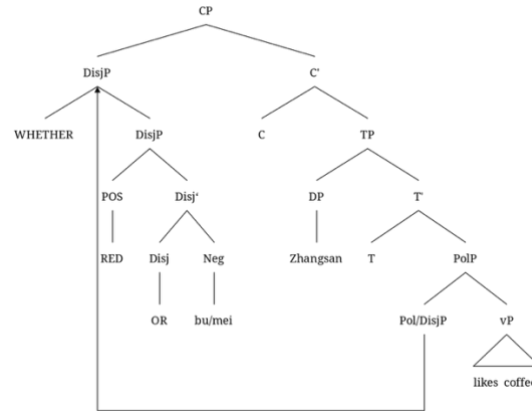
Zhangsan KE like drink coffee more good NE

These facts are based on C.-T. J. Huang’s (1991) and C.-T. J. Huang et al.’s (2009) observations about A-not-A, which led them to propose that the A-not-A morpheme is an interrogative operator that moves covertly to Spec, CP. We adopt their analysis of covert movement, but in our analysis, the A-not part and *shi-fou* are polar DisjP. What actually moves is the covert ‘whether’ that is adjacent to the polar DisjP; because its movement is covert, we do not know whether it covertly pied-pipes the polar DisjP.²⁵ (91) and the

²⁵ Since this ‘whether’ is covert, an alternative analysis is that it moves overtly as ‘whether’ does in Yixing, except that ‘whether’ does not pied-pipe the polar DisjP in A-not-A and *a/ke*-questions. While this analysis

following are the trees where the covert ‘whether’ moves and pied-pipes the DisjP covertly.

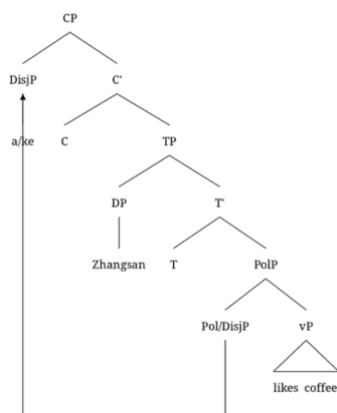
(102)



The same analysis of covert movement applies to *a/ke*-questions. Section 4 will show that *a* and *ke* realise the entire DisjP, including ‘whether’. Since they include the *wh*-feature, they undergo covert-movement to Spec, CP:

(103)

is in principle possible given the examples we have seen so far, section 4 will discuss colloquial Singapore Teochew, Dongliu and Hefei where *ke* clearly has the *wh*-feature. This *wh*-element stays in-situ and sentence-medially, but is subject to island effects, suggesting that ‘whether’ moves covertly rather than overtly.



To summarise, after section 2 discussed the first parameter of variation in yes/no questions (i.e., whether a yes/no question involves a polar DisjP), this section has introduced the second parameter—in ALTQVNS, whether ‘whether’ moves overtly or covertly. We use three new yes/no question constructions, A-not-A questions, *shifou*- and *a/ke*-questions, to show that while they involve a DisjP, they involve covert *wh*-movement which can move long-distance, but not across an island. This contrasts with Yixing *fe/me*-questions, which involve overt *wh*-movement. We agree with C.-T. J. Huang (1991), C.-T. J. Huang et al. (2009) and Holmberg (2016) that A-not-A and *a/ke*-questions involve covert *wh*-movement, but differ in the internal structure of A-not-A, and what is the *wh*-element.

4. Whether the DisjP is realised as a single VI or multiple ones

This section discusses the third parameter of variation across the Chinese varieties—in ALTQVNS, whether the polar DisjP is realised as a single VI or multiple ones. As we have claimed in the previous section, the DisjP is spelt out as multiple VIs in A-not-A and *shifou*- questions. They will be contrasted with *a* and *ke* in subsection 4.1, which realise the DisjP in a single VI. In subsection 4.2, we discuss the fact that the interaction of the

second and third binary parameters predicts there to be a type of yes/no question with overt *wh*-movement and realisation of the DisjP as a single VI. Then we show that *bo*-questions in Liangshan are precisely this.

4.1. *A* and *ke* realise the DisjP as a single VI

The previous sections have shown that *fe/me*-, A-not-A, *shifou*- and *a/ke*-questions all involve a polar DisjP. This DisjP is spelt out as multiple VIs, including some silent ones in *fe/me*-, A-not-A and *shifou*- questions. We think *a* and *ke* are most likely to be the DisjP in *a/ke*-questions, and the question is whether they are part of a multi-VI sequence including silent VIs (as in *fe/me*-questions), or they are a single VI. We argue for the latter because unlike the negation in A-not-A, *shifou*- and Yixing *fe/me*-questions, the morphology of *a* and *ke* is not sensitive to the aspect of the sentence. They remain *a* and *ke*, regardless of whether the sentence aspect is perfective (104b) or not (104a).

(104) a. li a/ke xihuan he kafei?

you A/KE like drink coffee

‘Do you like drinking coffee?’

b. li shangwu a/ke he-le na-bei kafei?

you morning A/KE drink-PERF that-cup coffee

‘Did you drink that cup of coffee this morning?’

We thus propose that the DisjP is realised as a single VI in *a/ke*-questions, specifically, the features of the DisjP are packaged into a single functional morpheme (i.e., the particle *a* or *ke*) in Late Insertion (see (109a-b) later).²⁶

Furthermore, we argue that the features expounded by *a* and *ke* also include the *wh*-feature of ‘whether’ in Spec, DisjP because *a* and *ke*’s counterpart in colloquial Singapore Teochew, *ka*, has been reported to be able to occur alone (105), or co-occur with A-not-A (106).²⁷

(105) Ah Meng ka suka ji bun zi?

Ah Meng KA like this CL book

‘Does Ah Meng like this book?’

(Cole and Lee 1997:190)

(106) Ah Meng ka su m suka ji bun zi?

Ah Meng KA RED NEG like this CL book

‘Does Ah Meng like this book?’

(Cole and Lee 1997:192)

²⁶ This is in line with Bobaljik’s (2012) analysis of portmanteaus, which are the result of Vocabulary Insertion ‘at a node dominating multiple features corresponding to multiple terminal nodes at some state of the derivation’ (Bobaljik 2012:148). Studies not specifically in the DM framework have also posited realising the features of an XP with a single VI (e.g., Neeleman and Szendrői 2007).

²⁷ Our Singapore Teochew speaker consultant found the co-occurrence of *ka* and A-not-A to be unnatural, suggesting there is inter-speaker variation in this.

We take this to mean that *ka* may either realise the entire polar DisjP, including the *wh*-element ‘whether’ as in (105), or only realise ‘whether’ as in (106).²⁸ In the latter case, the A-not-A form is the rest of the DisjP.²⁹ In both cases, *ka* realises the *wh*-feature.

Dongliu and Hefei have been reported to allow *ke* to co-occur with polar DisjP as well:

(107) *ke xiang bu xiang?* *Dongliu*

KE fragrant NEG fragrant

‘Does it smell good?’

(108) *ke na dong na bu dong?* *Hefei*

KE carry move carry NEG move

‘Can you carry it?’ (Zhu 1985:13)

Thus, we assume that *a* and *ke* also realise the *wh*-feature. As far as we know, *a* and *ke* cannot co-occur with A-not-A with reduplication of the first syllable in all the varieties

²⁸ This is a simplification of Cole and Lee’s (1997) analysis that *ka* is Q^0 , and there’s a covert *wh*-element in Spec, *ka*P that moves covertly.

²⁹ We think that the so-called A-not-A form in colloquial Singapore Teochew is not equivalent to the A-not-A form in Mandarin discussed in section 3. According to Cole and Lee (1997), the A-not-A form in colloquial Singapore Teochew can appear alone without *ka*; and when it does, it can be embedded in islands and still give rise to the relevant yes/no reading, in contrast to *ka*. We observe that this parallels the island insensitivity of alternative questions with transparent disjunction by *haishi* ‘or_{whether}’ in Mandarin (footnote 10), and thus think the A-not-A form in colloquial Singapore Teochew may be a transparent DisjP, in line with a similar point by Cole and Lee. In contrast, the island sensitivity of Mandarin A-not-A questions suggests that they involve covert *wh*-movement.

discussed in this paper (C.-T. J. Huang 1991).³⁰ This was confirmed by our consultants, suggesting that A-not-A with reduplication of the first syllable must include the *wh*-feature, and thus cannot co-occur with *a* and *ke*, which also expone the *wh*-feature.

It is worth mentioning that *ka* constitutes an indirect argument that *a* and *ke* move covertly. As Cole and Lee argued, *ka* undergoes covert *wh*-movement because when it is embedded in an island, we lose the relevant yes/no reading. If *ka*'s counterparts *a* and *ke* are single VIs that include the *wh*-feature, then their sentence-medial position and sensitivity to islands suggest that they also move covertly like *ka*.

In contrast to *a/ke*-questions, in *fe/me*-, A-not-A and *shifou*-questions, the polar DisjP is phonologically realised with distinct VIs. We elaborate the trees in (92)–(94) further in (109) to which we add *a/ke*-questions. As will be discussed in detail in section 5, this variation in (109) is consonant with some versions of the Borer-Chomsky Conjecture (BCC) that attribute crosslinguistic variation to different ways to realise the same features morphophonologically (e.g., Borer 2013).³¹

(109) *How polar DisjP is realised*

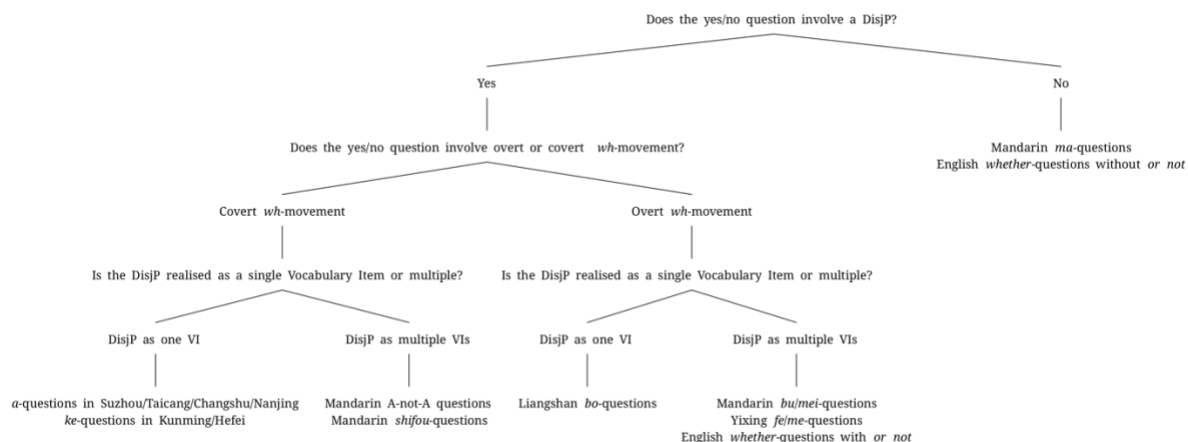
³⁰ Under this view, (107)–(108) would involve disjunction of the entire VP (i.e., C.-T. J. Huang's 1991 [AB not AB] forms), and the first disjunct is not just reduplication of the first syllable. This type of VP-disjunction does not need to include the *wh*-feature, and can thus co-occur with *ke*.

³¹ DM-based studies do not explicitly discuss BCC or deep reasons behind variation, but we think the DM tradition is consistent with it. As Embick (2015:32) proposed, 'languages may differ in terms of how they package their features into functional morphemes.' Here, Embick's feature packaging precisely concerns what feature(s) can be realised by a single functional item.

<i>a.</i>	<i>a</i>	<i>Nanjing, etc.</i>
<i>b.</i>	<i>ke</i>	<i>Hefei, etc.</i>
<i>c.</i>	<WHETHER> <POS> <OR> NEG	<i>Yixing</i>
<i>d.</i>	<WHETHER> RED <OR> NEG	<i>Mandarin A-not-A</i>
<i>e.</i>	<WHETHER> POS <OR> NEG	<i>Mandarin shifou</i>
<i>f.</i>	whether <POS> or not	<i>English</i>

Following is a more complete picture of parametric variation, repeated from (9), including all three parameters—(a) whether the yes/no question involves a DisjP; (b) whether the yes/no question involves overt or covert *wh*-movement; and (c) how the DisjP is realised morphologically.

(9) Parameters of Yes/no Questions



The second parameter, whether a yes/no question involves overt or covert *wh*-movement, is logically independent from the third parameter, how the DisjP is realised morphologically. Among covert movement constructions, we have seen variation along the third parameter: Mandarin A-not-A and *shifou*-questions realise the DisjP as multiple

VIs, whereas *a/ke*-questions realise the DisjP as a single one. We should expect this variation among overt movement constructions as well, where some realise the DisjP as multiple morphemes, as in Yixing *fe/me*-questions, while others realise it as a single VI that undergoes overt movement. We discuss precisely this possibility in the next subsection.

4.2. DisjP as a VI plus overt *wh*-movement: The case of Liangshan

Bo-questions in Liangshan variety of Southwestern Mandarin, spoken in Liangshan area of China's Sichuan Province, involve overt *wh*-movement and realisation of DisjP as a single VI. To the best of our knowledge, *bo*-questions have never been discussed in the literature before.

These questions end with the SFP *bo*, which is phonologically similar to the non-perfective negation marker *bu* in this variety. Given this resemblance, it may be tempting to analyse this question form in parallel with Yixing *fe/me*-questions or Mandarin *bu/mei*-questions, whose SFPs are negation. However, there is evidence that *bo* is not negation. As in Mandarin and Yixing, sentence negation in Liangshan is sensitive to the aspect: *bu* is used with the imperfective aspect (110a), and *mei* is used in perfective situations (110b).

(110) a. Zhangsan bu xihuan he kafei.

Zhangsan NEG like drink coffee

‘Zhangsan does not like to drink coffee.’

b. Zhangsan mei he kafei.

Zhangsan NEG drink coffee

‘Zhangsan didn’t drink coffee.’

In contrast, the yes/no question marker *bo* is invariant, regardless of aspect:

(111) a. Zhangsan xihuan he kafei bo?

Zhangsan like drink coffee BO

‘Does Zhangsan like to drink coffee?’

b. Zhangsan he-le kafei bo?

Zhangsan drink-PERF coffee BO?

‘Has Zhangsan drunk coffee?’

It may thus appear that *bo* is like *ma* in Mandarin—they both mark a yes/no question in the sentence-final position, and do not change their form based on aspect. However, when we apply the four tests of polar DisjP, *bo*-questions in Liangshan pattern do involve a DisjP.

First, they reject the answer *shi de* (‘It is so’):

(112) A: Zhangsan xihuan he kafei bo?

Zhangsan like drink coffee BO

‘Does Zhangsan like to drink coffee?’

B: #Shi de.

COP DE.

‘It is so.’

Second, *bo*-questions can generally be embedded under verbs like ‘know’, but cannot be embedded under ‘doubt’:

(113) a. wo bu xiaodei Zhangsan xihuan he kafei bo.

I NEG know Zhangsan like drink coffee BO

‘I don’t know whether Zhangsan likes to drink coffee.’

b. #Lisi huaiyi Zhangsan xihuan he kafei bo.

Lisi suspect Zhangsan like drink coffee BO

Intended: ‘Lisi suspects whether Zhangsan likes to drink coffee.’

Third, *bo*-questions can occur in unconditionals:

(114) buguan ni mingtian lai wo jia chi fan bo,

no.matter you tomorrow come my home have meal BO

wo jintian dou hui mai haoduo cai.

I today DOU will buy many groceries

‘Whether you’re coming for a meal, I’ll buy a lot of groceries today anyway.’

Finally, *bo*-questions do not allow negative propositions:

(115) *Zhangsan bu xihuan he kafei bo?

Zhangsan NEG like drink coffee BO

Intended: ‘Does Zhangsan not like to drink coffee?’

The empirical data suggest that *bo*-questions behave like Mandarin *ma*-questions in some ways but like Yixing *fe/me*-questions in others. First, (110) suggests that Liangshan *bo*, like Mandarin *ma* but unlike Yixing *fe/me*, is not negation because it is not sensitive to aspect. Then (112)–(115) suggest that *bo*-questions, like *fe/me*-questions but unlike *ma*-questions, involve a DisjP. Their general embeddability suggests that *bo*-questions do not involve SA⁰. Taken together, these facts lead us to propose that the polar DisjP in *bo*-

questions is realised as a single VI *bo*, just like *a* and *ke*. Since it realises the entire DisjP, not just the negative polarity in it, it is not sensitive to aspect.

Bo's sentence-final position, we argue, is due to the overt *wh*-movement of *wh*-DisjP. Sentence-final *bo* can give rise to a matrix yes/no question with embedded disjunctive choice. (116) has both the embedded reading, which can be answered by 'He claims that he doesn't', and the matrix reading, which can be answered by 'He doesn't claim that he does.'

(116) Zhangsan xuancheng ta xihuan kafei bo?

Zhangsan claim he like coffee BO

✓Embedded reading: 'Does Zhangsan claim that he likes coffee, or does Zhangsan claim that he doesn't like coffee?'

✓Matrix reading: 'Does Zhangsan claim that he likes coffee, or does Zhangsan not claim that he likes coffee?'

When embedded in a relative clause island, the embedded reading disappears, and only the matrix reading is possible. (117) can only be answered with 'He met such a student,' or 'He didn't meet such a student,' answers to the matrix reading. Its no-answer cannot be 'He met a student who doesn't like coffee.'

(117) Zhangsan renshi-le xihuan kafei de xuesheng bo?

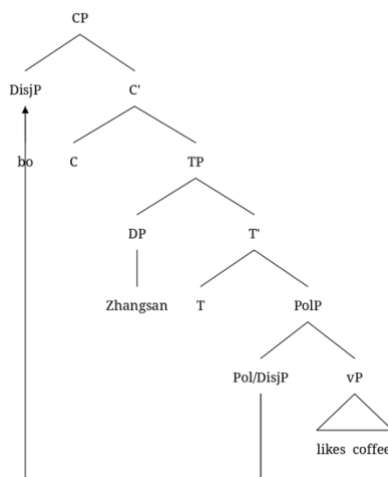
Zhangsan know-PERF like coffee DE student BO

*Embedded reading: 'Did Zhangsan meet a student who likes coffee, or did he meet a student who doesn't like coffee?'

✓Matrix reading: ‘Did Zhangsan meet a student who likes coffee, or did he not meet a student who likes coffee?’

Our syntactic analysis of *bo* is summarised by the tree in (118).

(118)



Liangshan thus completes the paradigm: *bo*-questions externalise the DisjP in a single VI as in Nanjing and Hefei *a/ke*-yes/no questions, but involve overt *wh*-movement as in Yixing *fe/me*-questions.

5. Conclusion

This paper has surveyed an apparently diverse set of ten yes/no question constructions across English and the Chinese variants, and argued that they are in fact complete systematic variation generated by just three binary parameters, which we summarise with the parameter hierarchy in (9). This apparent diversity of yes/no questions is due to their unique intersection of alternatives and question formation—on the one hand, questions may be formed with or without syntactic generation of alternatives; on the other hand,

questions may vary in how those alternatives are morphologically realised, and whether they involve overt or covert movement.

As we lay out these three parameters, we have made two main theoretical contributions. The first is for question semantics and typology. We have provided morphological support for two distinct interrogative C^0 s, one in ALTQVNs and the other in POLQs. Not only are they realised differently, but we have also shown that they are associated with different syntactic properties, where ALTQVN C^0 requires (overt or covert) *wh*-movement, but POLQ C^0 doesn't. The fact that POLQs do not involve any *wh*-movement or even genuine *wh*-in-situ enriches the typology of questions and suggests that there are at least three question types across languages.

Our second contribution is for the theories of parametric variation. Many different but interrelated approaches to parametric variation have been proposed in the Minimalist Program (for a recent comprehensive study, see Roberts 2019). We divide them into three types—two different interpretations of 'Borer-Chomsky Conjecture (BCC)', and what we call the 'Chomskyan Externalisation'. We describe each approach in more detail below, and argue that the intricate cross-linguistic and cross-dialectal variation uncovered in this paper supports all three approaches, with each parameter being accounted for by one approach.

Perhaps the best-known approach to parametric variation in minimalism is BCC (e.g., Baker 2008; Roberts 2019), which attributed variation to the lexicon. Less explicitly stated in the literature is the fact that there are actually two different versions of BCC. The first is what we call the 'Feature Availability' version, where cross-linguistic

variation is the result of the presence or absence of certain features in the lexicon. Specifically, Chomsky (2001) argued that variation occurs because a language makes a one-time selection of the universal set of formal features before it builds up its lexicon, and different languages may choose different sets of features. This idea was supported using a wide range of cross-linguistic data in major parametric studies such as Biberauer et al. (2014) and Roberts (2019). Following the ‘Feature Availability’ version of the BCC, our first parameter (i.e., whether a language has POLQ or ALTQVN or both) is due to what features a language has chosen for its C^0 .

The second version of BCC is Borer’s (2013), which focused on morphophonological realisation of features. She attributed cross-linguistic variation to what she called ‘realisational properties’, where the same functional item or feature might be realised by different surface phonological forms (Borer 2013:630). We call this the ‘Feature Realisation’ version of the BCC, to be contrasted with the more widely adopted ‘Feature Availability’ version that we introduced above. Our third parameter (i.e., whether the DisjP is realised as a single VI or multiple ones) is precisely variation in features’ morphophonological realisation and thus supports the ‘Feature Realisation’ version of the BCC.

There is a series of recent works under the ‘Chomskyan Externalisation’ approach that argued that cross-linguistic variation can arise due to externalisation (Berwick and Chomsky 2016; Chomsky 2019; Chomsky et al. 2023): there is no variation in the Merge-based narrow syntax computation, and variation can arise when the universal hierarchical structure is externalised at the sensorimotor (SM) interface. In some way, the

‘Feature Realisation’ version of the BCC may be subsumed under this approach because morphophonological realisation is a part of externalisation. But the works in the ‘Chomskyan Externalisation’ approach have primarily focused on another aspect of externalisation, linearisation, such as head directionality and copy spell-out. Our second parameter (i.e., whether a yes/no question involves overt or covert *wh*-movement) is precisely variation in linearisation—which *wh*-copy in the *wh*-movement chain is pronounced. Furthermore, we would like to argue that the ‘Feature Realisation’ version of BCC and the Chomskyan Externalisation not only share the same core that variation is restricted to externalisation, but also complement each other: with the former approach focusing on the lexicon, and the latter on word order, together they provide a full picture of how externalisation may lead to parametric variation. This is in line with Chomsky’s (2019:59) claim that ‘parameter setting is restricted to externalization and the lexicon.’

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